

**Frankfurt University
of Applied Sciences**

– Faculty of Computer Science and Engineering –

**Conformal Uncertainty for Operator Deep
Smoothing of Implied-Volatility Surfaces**

Thesis submitted in order to obtain the academic degree
Master of Science (M.Sc.)

submitted on

Aniket Prakash Nighot

Student ID: 1470888

First Supervisor : Dr. Martin Simon
Second Supervisor : Dr. Doina Logofatu

Declaration

Hereby I assure that I have written the presented thesis independently and without any third party help and that I have not used any other tools or resources than the ones mentioned/referred to in the thesis.

Any parts of the thesis that have been taken from other published or yet unpublished works in terms of their wording or meaning are thoroughly marked with an indication of the source.

All figures in this thesis have been drawn by myself or are clearly attributed with a reference.

This work has not been published or presented to any other examination authority before. I am aware of the importance of the affidavit and the consequences under examination law as well as the criminal consequences of an incorrect or incomplete affidavit.

Frankfurt,

Aniket Prakash Nighot

Abstract

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Prof. Martin Simon, for his continuous support, guidance, and insightful feedback throughout the course of this thesis. His expertise in quantitative modeling and encouragement to explore unconventional approaches have been invaluable to the development of this work.

I am deeply thankful to the faculty members of the Master's programme in *High Integrity Systems* at Frankfurt University of Applied Sciences for providing a rigorous academic environment that shaped both my technical foundations and critical thinking.

To my friends and classmates, thank you for the camaraderie, discussions, and the shared journey over the past semesters.

Lastly, I am deeply grateful to my family for their unwavering love, patience, and encouragement. Without their constant support and belief in me, this thesis would not have been possible.

Contents

Abstract	i
List of Figures	vi
List of Tables	vii
1 Introduction	1
1.1 Motivation and Problem Statement	2
1.2 Research Objectives	2
1.3 Contributions	3
1.4 Thesis Outline	3
2 Background and Related Work	4
2.1 Implied Volatility and the Options Market	4
2.1.1 European Options and the Black-Scholes Model	4
2.1.2 Implied Volatility	4
2.1.3 The Implied Volatility Surface	5
2.1.4 No-Arbitrage Conditions	5
2.2 Implied Volatility Smoothing	5
2.2.1 Parametric Approaches	5
2.2.2 Neural Network-Based Approaches	6
2.2.3 Operator Deep Smoothing	6
2.3 Conformal Prediction	6
2.3.1 Foundations	7
2.3.2 Inductive Conformal Prediction	7
2.3.3 Nonconformity Scores	7
2.3.4 Mondrian Conformal Prediction	7
2.3.5 Conformal Prediction for Time Series	8
2.4 Uncertainty Quantification in Financial Machine Learning	8
2.5 Summary	8
3 Methodology	9
3.1 Problem Setup and Notation	9

3.2	Nonconformity Scores	10
3.2.1	Score A: Spread-Normalised Price Error	10
3.2.2	Score B: Vega-Normalised Implied Volatility Error	10
3.3	Rolling Window Calibration and Mondrian Binning	10
3.4	Prediction Bands	11
3.4.1	Price-Spread Mode	11
3.4.2	Vega-IV Mode	11
3.5	No-Arbitrage Projection	12
3.5.1	Calendar and Butterfly Constraints	12
3.5.2	Projection Algorithm	12
3.5.3	Coverage after Projection	13
3.6	Evaluation Metrics	14
3.7	Summary	14
4	Implementation	15
4.1	Software Architecture	15
4.2	Nonconformity Score Computation	15
4.3	Rolling Window Calibration	16
4.4	Mondrian Spatial Binning	16
4.5	Prediction Band Construction	17
4.6	No-Arbitrage Projection	17
4.7	Data Acquisition and Preprocessing	18
4.8	Experiment Notebook	18
5	Experiments and Results	20
5.1	Experimental Setup	20
5.2	Calibration Behaviour	21
5.2.1	Rolling Window Warm-up	21
5.2.2	Global versus Mondrian Quantiles	21
5.3	No-Arbitrage Projection	22
5.3.1	Aggregate Violation Counts	22
5.3.2	Score B: Elimination of Violations	24
5.3.3	Score A: Partial Violation Reduction	24
5.4	Nonconformity Score Comparison	24
5.4.1	Score Magnitude and Quantile Scale	24
5.4.2	Band Width	24
5.4.3	Coverage Stability	25
5.4.4	Summary of Score Comparison	27
5.5	Coverage Results	27
5.5.1	Per-Surface Coverage	27

5.5.2	Contribution of Mondrian Spatial Binning	28
5.6	Sharpness	28
5.6.1	Overall Band Width	28
5.6.2	Spatial Variation of Band Width	28
5.6.3	Sharpness Relative to Score A	29
5.7	Summary	29
6	Discussion	30
6.1	The Exchangeability Assumption in Practice	30
6.1.1	Serial Dependence and the Rolling Window	30
6.1.2	Temporal Variation in Coverage	30
6.1.3	Contribution of Mondrian Binning	31
6.2	Practical Implications	31
6.3	Limitations	32
6.4	Future Research	32
6.5	Summary	32
7	Conclusion	34
	Bibliography	35

List of Figures

4.1	SPX options dataset: option quote count per surface (top) and collection timeline (bottom) across 56 successfully collected surfaces from December 2025 to August 2026. Blue indicates calibration surfaces (indices 0–14); red indicates test surfaces (indices 15–55).	19
5.1	Score B global quantile $\hat{q}$, Mondrian quantile range, and per-surface mean score across the 15 calibration surfaces.	22
5.2	Calendar condition in selected slices of the lower Score B band for surface 15, before and after projection.	23
5.3	Butterfly condition in selected maturity slices of the lower Score B band for surface 15, before and after projection.	23
5.4	Post-projection per-surface Mondrian coverage for Score A and Score B across the 41 evaluation surfaces.	26
5.5	Post-projection mean IV band width for Score A and Score B across the 41 evaluation surfaces. The two vertical axes show the different width scales of the two scores.	26

List of Tables

4.1	Modules of the volatility_smoothing/conformal/ package and their roles.	15
4.2	Hyperparameters used in all experiments.	18
5.1	Score B calibration quantiles across 15 surfaces.	22
5.2	Combined calendar and butterfly violation counts on the lower and upper Mondrian bands across 41 walk-forward evaluation surfaces.	23
5.3	Average per-surface mean and median IV band widths across the 41 evaluation surfaces.	24
5.4	Post-projection per-surface Mondrian coverage for Score A and Score B across all 41 evaluation surfaces. Target: $1 - \alpha = 90\%$	25
5.5	Matched post-projection Score B coverage on all 41 evaluation surfaces ($1 - \alpha = 90\%$).	27

Chapter 1

Introduction

Continued growth in options trading has increased the practical importance of reliable methods for derivative pricing and risk management. Cboe Global Markets reported record trading volumes in 2025, with 4.6 billion contracts traded across its four options exchanges, marking the sixth consecutive annual record. S&P 500 Index (SPX) options accounted for 970.6 million contracts and an average daily volume of 3.9 million contracts [7]. This level of activity strengthens the need for efficient methods that construct implied volatility (IV) surfaces from complex and irregular market data.

Implied volatility converts a quoted option price into the volatility parameter that reproduces the price within the Black–Scholes model. The implied volatility surface collects these values across strikes and maturities at a given time and serves as an important input for derivative pricing, hedging, and risk management. Constructing a smooth and financially consistent surface from the discrete and noisy quotes available in practice is a central challenge in quantitative finance [10].

Parametric methods address this problem by fitting a functional form to each new set of market quotes. The widely used Stochastic Volatility Inspired (SVI) model [9], for example, parameterises each maturity slice using a five-parameter form. Such calibration approaches generally require numerical optimisation for each market snapshot and may be sensitive to the selected initial values [10].

Recent advances in machine learning have opened new possibilities for volatility surface construction. Gonon, Jacquier, and Wiedemann [10] propose an Operator Deep Smoothing approach based on a Graph Neural Operator (GNO). The model accepts market quotes at the strike–maturity locations available in a given snapshot and produces a smooth implied volatility surface in a single forward pass. Its training objective encourages adherence to the relevant financial constraints. Because the architecture can process changing input configurations, it does not require a separate numerical calibration for every market snapshot or an initial interpolation of the quotes onto a fixed grid.

The GNO produces a deterministic smoothed surface but does not by itself provide prediction intervals around that surface. An uncertainty layer can therefore complement the point prediction by showing how closely the observed implied volatilities are expected to lie around the fitted surface. This is particularly useful when the density and quality of the available quotes vary across the surface or over time.

Common uncertainty quantification approaches add computational requirements to the underlying model. Monte Carlo dropout uses repeated stochastic forward passes as an approximation to Bayesian inference [8], while deep ensembles require multiple models to be trained and evaluated [11]. Conformal prediction instead post-processes the output of a fixed predictor. Under exchangeability of the calibration and test observations, standard conformal methods provide a finite-sample marginal coverage guarantee without requiring a parametric model for the prediction errors [19, 2]. This makes conformal prediction a suitable uncertainty layer around the pre-trained GNO. Since the empirical application in this thesis is temporally ordered, the rolling implementation and its achieved coverage are evaluated directly rather than treating exchangeability as exact.

A key challenge in applying conformal prediction to implied volatility surfaces is the design of suitable nonconformity

scores. An unscaled residual treats errors of the same numerical size equally across the surface, although quote spreads and option sensitivities vary with strike and maturity. This thesis therefore studies two economically motivated scores based on bid–ask spreads and option Vega. These quantities scale the residuals using information available for each option and allow the resulting band widths to vary across the surface.

1.1 Motivation and Problem Statement

The central problem addressed in this thesis is the construction of calibrated uncertainty bands for implied volatility surfaces produced by the Graph Neural Operator approach. More specifically, this work develops an efficient uncertainty quantification layer that:

1. Targets finite-sample marginal coverage across observed surface points under exchangeability, without requiring a parametric model for the prediction errors;
2. Accounts for heteroskedastic prediction errors through economically motivated normalization;
3. Responds to changing market conditions through rolling calibration windows;
4. Produces uncertainty bands in both implied volatility and price spaces; and
5. Applies a post-processing routine that enforces calendar and butterfly conditions on the lower and upper band surfaces on a regular grid.

Options data are heterogeneous across both time and the strike–expiry domain. Liquidity, quote density, bid–ask spreads, and price sensitivity vary between regions of the surface. The proposed framework therefore allows the band width to vary across observed points and evaluates whether spatially adaptive calibration improves coverage across the surface while retaining marginal evaluation over all observations.

1.2 Research Objectives

This thesis pursues the following specific objectives:

1. **Design economically motivated nonconformity scores:** Develop two heteroskedastic scoring mechanisms that normalize prediction errors by (a) bid–ask spreads in price space and (b) Vega-weighted sensitivity in implied volatility space, ensuring uncertainty bands align with trading frictions and option Greeks.
2. **Implement rolling window calibration:** Establish a temporal adaptation mechanism using a sliding window of recent market snapshots to compute conformal quantiles, enabling the framework to track regime changes in market volatility dynamics.
3. **Evaluate spatial heterogeneity via Mondrian conformal prediction:** Investigate whether partitioning the strike–expiry domain into spatial bins and computing region-specific quantiles improves coverage uniformity across areas with varying liquidity.
4. **Construct prediction bands in dual spaces:** Provide practitioners with uncertainty estimates in both implied volatility space (for volatility trading) and price space (for delta hedging and risk management).
5. **Apply calendar and butterfly adjustments to prediction bands:** Apply an iterative projection algorithm to the lower and upper band surfaces and evaluate the resulting calendar and butterfly conditions on the finite (ρ, z) grid used by the implementation.
6. **Validate empirical coverage:** Assess whether the conformal band surfaces achieve the nominal marginal coverage probability $(1 - \alpha)$ across observed quote locations in S&P 500 options data spanning December 2025 to August 2026.

1.3 Contributions

The main contributions of this thesis are:

1. **First application of conformal prediction to neural operator-based volatility surfaces:** While conformal prediction has been applied to operator learning in scientific computing [14], and to implied volatility forecasting without neural operators [6], to the best of our knowledge this work represents the first integration of distribution-free uncertainty quantification with operator learning architectures specifically in the context of implied volatility smoothing.
2. **Economically aligned nonconformity scores:** Adaptation of spread-normalized price errors and Vega-normalized implied volatility errors for conformal calibration around the GNO surface. These constructions incorporate bid-ask spread information and option sensitivity when determining the band widths.
3. **Temporal adaptation via rolling calibration:** Use of a rolling window of recent calibration data so that the conformal quantiles respond to changes in the observed prediction errors. Its performance on the temporally ordered surfaces is assessed empirically through the walk-forward evaluation.
4. **Grid-based projection of uncertainty bands:** Implementation of an iterative procedure that applies the discrete calendar and butterfly adjustments used in this thesis to both the lower and upper band surfaces, together with diagnostics for residual violations and lower-upper crossings.
5. **Empirical validation on real market data:** Comprehensive evaluation on S&P 500 options data spanning December 2025 to August 2026, providing evidence that conformal prediction can deliver empirically calibrated uncertainty surfaces in a challenging real-world financial application.

1.4 Thesis Outline

The remainder of this thesis is structured as follows:

Chapter 2: Background and Related Work provides the necessary background on implied volatility surfaces, no-arbitrage constraints, traditional smoothing methods, neural operators, and the theoretical foundations of conformal prediction. We review related work on uncertainty quantification in financial machine learning and position our contribution within this literature.

Chapter 3: Methodology presents the technical details of our conformal uncertainty quantification framework. We formally define the problem, introduce the two nonconformity scores (spread-normalized and Vega-normalized), describe the rolling window calibration mechanism, explain Mondrian conformal prediction for spatial heterogeneity, detail the construction of prediction bands in both price and implied volatility spaces, and present the grid-based projection procedure used to apply calendar and butterfly adjustments to the resulting band surfaces.

Chapter 4: Implementation describes the software architecture, data preprocessing pipeline, and integration with the pre-trained GNO model from Gonon et al.

Chapter 5: Experiments and Results reports the empirical evaluation on S&P 500 options data. We present coverage and sharpness results, visualizations of the uncertainty bands, the comparison of Score A and Score B, the comparison of global and Mondrian calibration, and the projection diagnostics.

Chapter 6: Discussion interprets the results, examines the practical implications for risk management and trading, discusses the limitations of the approach (particularly regarding the exchangeability assumption), and considers the financial validity of the uncertainty bands.

Chapter 7: Conclusion summarizes the contributions, reflects on the broader significance of distribution-free uncertainty quantification for financial machine learning, and outlines directions for future work, including extensions to online conformal prediction, adaptive spatial binning, and integration with richer option pricing models.

Chapter 2

Background and Related Work

This chapter establishes the technical foundations on which the thesis builds. We begin by introducing the financial setting of implied volatility surfaces and the no-arbitrage conditions they must satisfy. We then survey classical and machine learning-based approaches to implied volatility smoothing, situating the Graph Neural Operator method of Gonon, Jacquier, and Wiedemann [10] within this landscape. Finally, we review the theoretical framework of conformal prediction and discuss related work on uncertainty quantification in financial machine learning.

2.1 Implied Volatility and the Options Market

2.1.1 European Options and the Black-Scholes Model

Following the notation of Gonon et al. [10], we consider a market of European options written on an underlying asset observed at a reference time T_0 . Each option is identified by its expiry $T \in (T_0, \infty)$ and its strike $K \in (0, \infty)$. The time-to-expiry is denoted $\tau = T - T_0$ and the log-moneyness is defined as $k = \log(K/F_{T_0,T})$, where $F_{T_0,T}$ denotes the time- T forward price of the underlying asset.

The Black-Scholes model [5] provides a closed-form expression for the forward-normalised price of a European call option with time-to-expiry τ and log-moneyness k , as a function of the volatility parameter $v \in (0, \infty)$:

$$\text{BS}(\tau, k, v) = \Phi(d_1(\tau, k, v)) - e^k \Phi(d_2(\tau, k, v)), \quad (2.1)$$

where Φ denotes the cumulative distribution function of the standard normal distribution and

$$d_1(\tau, k, v) = \frac{-k}{v\sqrt{\tau}} + \frac{1}{2}v\sqrt{\tau}, \quad d_2(\tau, k, v) = \frac{-k}{v\sqrt{\tau}} - \frac{1}{2}v\sqrt{\tau}. \quad (2.2)$$

The sensitivity of this forward-normalised option price with respect to the volatility parameter, known as Vega, plays a central role in this thesis and is given by:

$$\mathcal{V}(\tau, k, v) = \varphi(d_1(\tau, k, v))\sqrt{\tau}, \quad (2.3)$$

where φ denotes the standard normal density [5].

2.1.2 Implied Volatility

While the Black-Scholes model does not fully describe observed market dynamics, its mathematical tractability makes it a widely used quoting convention. Let $C(T, K)$ denote the undiscounted market price of a call option. Its implied volatility $v(\tau, k)$ is defined as the volatility value that satisfies

$$C(T, K) = F_{T_0,T} \cdot \text{BS}(\tau, k, v(\tau, k)). \quad (2.4)$$

Implied volatility therefore expresses an observed option price as a market-implied Black–Scholes volatility parameter. This provides a common quoting scale for comparing the relative expensiveness of options across strikes, maturities, underlyings, and interest-rate environments [10].

2.1.3 The Implied Volatility Surface

At any given instant, an options market offers contracts across a range of strikes and maturities. Inverting the Black–Scholes formula [5] for each quoted option price yields a corresponding implied volatility. The collection of these values, viewed as a function of time-to-expiry τ and log-moneyness k [9], defines the implied volatility surface. This surface provides a three-dimensional snapshot of how the market prices uncertainty across strikes and maturities.

The implied volatility surface exhibits two well-documented structural features [9]. Across strikes at a fixed maturity, implied volatility typically forms a skewed smile. Across maturities, it displays a term structure that rises or falls with prevailing market conditions. These features summarise how observed option prices vary across the strike–expiry domain and make the surface an important input for derivative pricing, hedging, and portfolio risk measurement.

Because options trade only at a discrete and irregularly spaced set of strikes and maturities, and because this set changes as contracts expire and new ones are listed, the implied volatility surface is not directly observable as a continuous object. It must instead be reconstructed from noisy market quotes through interpolation, extrapolation, and smoothing. Constructing and quantifying the uncertainty of this reconstruction is the central problem this thesis addresses.

2.1.4 No-Arbitrage Conditions

Not every smooth surface defined over the strike–expiry domain corresponds to a financially consistent option-price surface. Static no-arbitrage requirements are commonly expressed through calendar and butterfly conditions, as formalised in Theorem 2.1 of Gonon et al. [10] and related to the SVI framework of Gatheral and Jacquier [9]:

- **No-calendar-arbitrage:** At fixed log-moneyness, total implied variance $\tau v(\tau, k)^2$ must not decrease as maturity increases. This condition corresponds to the required ordering of option prices across maturities [10].
- **No-butterfly-arbitrage:** At each fixed maturity, the corresponding call-option prices must be convex in strike. In implied-volatility terms, this is expressed through a condition involving the smile and its strike derivatives and ensures that the associated risk-neutral density remains non-negative [10].

These conditions provide the financial basis for the post-processing routine used in this thesis. On a regular grid, the routine applies the corresponding discrete calendar and butterfly conditions to the lower and upper band surfaces. The analytic conditions and their implementation are discussed in Chapter 3.

2.2 Implied Volatility Smoothing

2.2.1 Parametric Approaches

The task of implied volatility smoothing refers to fitting a smooth, arbitrage-free surface $\hat{v}$ to a finite collection $\mathbf{v} = \{v(\tau_i, k_i)\}_{i=1}^P$ of observed implied volatilities [10]. Direct interpolation schemes such as cubic splines can produce smooth fits, but they do not automatically enforce the calendar and butterfly conditions required across the surface.

The industry-standard approach is the Stochastic Volatility Inspired (SVI) model [9], which parameterises total implied variance at each maturity using a five-parameter functional form that captures the characteristic skew and smile. Its extension to full surfaces through the Surface SVI (SSVI) parameterisation [9] provides tractable conditions under which the resulting surface is free of static arbitrage. SVI and its variants have been widely adopted by practitioners.

Despite their practical success, parametric models require a new numerical optimisation whenever the set of market quotes is updated. The resulting calibration depends on the chosen parameterisation, objective function, and optimisation procedure [10].

2.2.2 Neural Network-Based Approaches

The limitations of parametric calibration have motivated a growing body of research on neural network-based implied volatility smoothing. Ackerer et al. [1] introduced deep neural network architectures combined with no-arbitrage soft constraints for volatility surface construction. Their framework also reports uncertainty estimates in regions with few or no observations, although it does not provide conformal coverage guarantees.

A related line of work applies variational autoencoders to volatility smoothing [4], exploiting the fixed-grid structure of FX options markets where strikes are quoted at standard deltas. As noted by Gonon et al. [10], this specificity limits direct application to equity index options such as SPX, where the set of available strikes and maturities changes continuously as contracts expire and new ones are listed.

A further limitation of classical neural network architectures in this setting is the requirement for fixed-size inputs [10]. Options data is inherently irregular: at each point in time, a different collection of strikes and maturities is available. Applying standard feedforward or convolutional networks therefore requires additional preprocessing and a fixed representation of the changing market data.

2.2.3 Operator Deep Smoothing

Gonon, Jacquier, and Wiedemann [10] address the irregular structure of options data by recasting implied volatility smoothing as an operator learning problem. Rather than fitting a parameterised function to each new set of market quotes, they train a map

$$F : \mathcal{A} \rightarrow \mathcal{U}$$

between function spaces, where the input $\mathcal{A}$ consists of collections of observed implied volatilities and the output $\mathcal{U}$ is the space of smooth surface functions. At inference time, this operator is evaluated in a single forward pass, bypassing the per-instance optimisation loop that parametric calibration requires.

The market observations are treated as point samples of a continuous volatility function $v : D \rightarrow \mathbb{R}$, where D is defined in the transformed coordinates

$$\rho = \sqrt{\tau}, \quad z = k/\rho,$$

giving $D = (0.01, 1) \times (-1.5, 0.5)$. These coordinates spread the data more uniformly across the input domain [10].

The architecture they adopt is the Graph Neural Operator (GNO) of Li et al. [13]. Its graph-based kernel aggregation can process observations on irregular and changing sets of input locations. Gonon et al. [10] make one structural modification by removing the pointwise linear transformation from the first GNO layer. This allows the first hidden state to be evaluated at output locations that need not coincide with the observed input locations. Together, the graph construction and this modification enable the model to map changing collections of quoted strikes and maturities to a smoothed surface without fixed-grid preprocessing.

The model is trained on nine years of intraday S&P 500 options data (2012–2020). On the held-out 2021 test year, it achieves substantially lower mean absolute percentage error than the SVI benchmark. The reported calendar and butterfly penalty terms are zero or effectively zero on average across the test set [10].

The trained model is deterministic and returns a single smoothed surface without accompanying prediction intervals. The present thesis complements this surface estimate with conformal uncertainty bands.

2.3 Conformal Prediction

2.3.1 Foundations

Conformal prediction was originally developed by Vovk, Gammerman, and Shafer; the framework is surveyed by Shafer and Vovk [19] and, more recently, by Angelopoulos and Bates [2]. Its central property is the ability to produce prediction sets with finite-sample marginal coverage guarantees without imposing any parametric assumptions on the data-generating process. The key condition required is *exchangeability*: the joint distribution of the calibration and test observations must remain unchanged when their indices are permuted [2]. This condition is natural for observations drawn from a common stable process, while temporally ordered financial data require additional care.

Following the split conformal construction described in Angelopoulos and Bates [2], the procedure works as follows. Given a pre-trained deterministic predictor and a held-out calibration dataset $\{(X_i, Y_i)\}_{i=1}^n$, a nonconformity score $A(X_i, Y_i)$ is computed for each calibration point, quantifying how poorly the predictor accounts for that observation. The $(1 - \alpha)$ -quantile of these n scores, corrected for the finite sample size, is

$$\hat{q}_{1-\alpha} = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n}(\{A(X_i, Y_i)\}_{i=1}^n). \quad (2.5)$$

For a new test input X_{n+1} , the prediction set is then

$$\mathcal{C}(X_{n+1}) = \{y : A(X_{n+1}, y) \leq \hat{q}_{1-\alpha}\}. \quad (2.6)$$

Under exchangeability of the $n + 1$ data points, this construction satisfies the marginal coverage guarantee

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha. \quad (2.7)$$

The guarantee is non-asymptotic: it holds for every finite n and for any choice of underlying predictor and nonconformity score, without requiring the model to be well-specified or calibrated in any distributional sense [19, 2]. Importantly, this is a marginal statement about an individual test observation. In the present application, the conformal intervals are evaluated over the strike-expiry domain and together form complete lower and upper band surfaces around the GNO estimate. The marginal guarantee applies to an individual evaluation location and does not require every location on a daily surface to be covered simultaneously.

2.3.2 Inductive Conformal Prediction

The computationally practical version of conformal prediction is the inductive (or split) conformal predictor [16, 12, 2]. It uses a held-out calibration set that is separate from the training data, computing nonconformity scores only once rather than retraining the predictor for each test point. This separation makes inductive conformal prediction directly applicable as a post-hoc wrapper around any pre-trained model, which is the approach adopted in this thesis.

2.3.3 Nonconformity Scores

The choice of nonconformity score determines the shape and adaptivity of the resulting prediction sets [2]. The simplest choice for regression is the absolute residual $A(x, y) = |\hat{f}(x) - y|$, which yields symmetric prediction intervals of constant width. More adaptive scores normalise the residual by an estimate of local uncertainty, producing intervals that are wider in regions where the model is less reliable. Conformalized Quantile Regression (CQR) [17] uses quantile regression estimates to produce locally adaptive prediction intervals with valid marginal coverage. The design of economically motivated nonconformity scores tailored to implied volatility is one of the central contributions of this thesis.

2.3.4 Mondrian Conformal Prediction

Mondrian conformal prediction [19, 2] extends the basic framework toward group-conditional coverage by partitioning the input space into disjoint bins and computing a separate conformal quantile within each bin. Under exchangeability, the unmodified Mondrian construction provides marginal coverage conditional on membership in a predefined bin. This is particularly relevant for implied volatility surfaces, where uncertainty varies systematically with liquidity across the strike-expiry domain. In this thesis, spatial binning is used primarily to create adaptive band widths across the surface, and its coverage behaviour is assessed empirically.

2.3.5 Conformal Prediction for Time Series

Applying conformal prediction to financial data requires particular care because financial time series exhibit serial dependence and changing market regimes. As a result, the relevance of older calibration observations can decrease over time. One practical response is to restrict calibration to a sliding window of the most recent observations, so that the conformal quantile reflects current rather than distant historical conditions [3, 2]. Although a rolling window does not by itself establish exchangeability, it makes the calibration responsive to evolving market conditions. Rolling window calibration is therefore used in this thesis, and its coverage in the time-dependent setting is assessed empirically.

2.4 Uncertainty Quantification in Financial Machine Learning

Predictive uncertainty is a broad concern in machine learning and is directly relevant to financial forecasting and risk management. General approaches include Monte Carlo dropout [8], which estimates uncertainty through repeated stochastic forward passes, and deep ensembles [11], which combine predictions from multiple independently trained networks. These methods provide useful uncertainty estimates but require either repeated model evaluations or the training of several models. Conformal prediction offers a model-agnostic alternative that calibrates prediction sets using held-out observations and, under exchangeability, provides finite-sample marginal coverage.

Research connected to options has considered neural networks in several related settings. Wiese et al. [21] use generative models to simulate equity option-market time series, while Ruf and Wang [18] study neural networks for option hedging. These applications differ from constructing uncertainty bands around a smoothed implied volatility surface. More directly, Ackerer et al. [1] construct arbitrage-aware neural-network implied volatility surfaces and quantify prediction uncertainty in sparsely observed regions, but do not use conformal calibration or neural operators. Canete [6] applies online conformal prediction by treating implied volatility as a point forecast of future realised volatility, rather than constructing bands around a smoothed implied volatility surface. Ma et al. [14] combine conformal prediction with operator learning for scientific and engineering problems, including Darcy flow and car-surface pressure prediction, but not for financial surfaces. Within the literature reviewed in this thesis, no previous study was identified that combines conformal prediction with neural-operator implied volatility smoothing. The present thesis investigates this specific combination.

2.5 Summary

This chapter has established the financial and statistical foundations of the thesis. The implied volatility surface is a central object in options markets and is shaped by calendar and butterfly conditions. Classical parametric methods calibrate this surface instance-by-instance, whereas the operator deep smoothing approach of Gonon et al. [10] learns a mapping from irregular market observations to a smooth surface that can be evaluated in a single forward pass. The resulting GNO estimate is deterministic, and this thesis complements it with conformal uncertainty bands. Under exchangeability, conformal prediction provides a distribution-free framework for constructing prediction sets with finite-sample marginal coverage. Evaluating the calibrated intervals over the strike-expiry domain produces complete lower and upper surfaces around the GNO estimate. Rolling window calibration makes the procedure responsive to changing market conditions, while Mondrian binning provides spatial adaptation across the surface. Their performance in the time-dependent financial setting is evaluated empirically. The following chapter presents the methodology that combines these elements into an uncertainty quantification framework for implied volatility surfaces.

Chapter 3

Methodology

This chapter presents the conformal uncertainty quantification framework developed in this thesis. Section 3.1 introduces the problem setup and notation. Section 3.2 defines the two nonconformity scores used to measure prediction error. Section 3.3 describes the rolling window calibration mechanism and Mondrian spatial binning that adapt the conformal quantile to temporal and spatial heterogeneity. Section 3.4 details the construction of prediction bands in both price and implied volatility space. Section 3.5 presents the no-arbitrage projection step that enforces the constraints introduced in Chapter 2 on the resulting bands. Section 3.6 defines the evaluation metrics used to assess the framework in Chapter 5.

3.1 Problem Setup and Notation

Following the transformed coordinate system introduced by Gonon et al. [10] and adopted throughout this thesis, an option quote is identified by a point $x = (\rho, z) \in D$, where $D = (0.01, 1) \times (-1.5, 0.5)$, $\rho = \sqrt{\tau}$ is the square root of time-to-expiry, and $z = k/\rho$ is the maturity-normalised log-moneyness. For a finite and irregular collection of observed quotes, the trained Graph Neural Operator $\hat{F}^\theta$ of Gonon et al. [10] produces a smooth implied volatility field $\hat{v} : D \rightarrow (0, \infty)$. The GNO is treated in this thesis as a fixed, pre-trained surface estimator. The conformal framework is added as a post-hoc uncertainty layer, without modifying the GNO architecture or its training procedure.

Two market quantities are used to define the nonconformity scores. Let $\text{BS}_\pm(x, v)$ denote the forward-normalised, sign-aware Black–Scholes price used by Gonon et al. [10]. The implementation converts this quantity to the corresponding option-price units through

$$P_{t,\pm}(x, v) = D_{t,T} F_{t,T} \text{BS}_\pm(x, v), \quad (3.1)$$

where $D_{t,T}$ and $F_{t,T}$ are the discount factor and forward price for the relevant maturity. The spread $s_t(x)$ is the observed ask price minus the bid price in the same units as $P_{t,\pm}$. Score B uses the forward-normalised Vega $\mathcal{V}(x, v) = \varphi(d_1(x, v))\rho$. These quantities are built from the Black–Scholes terms introduced in Section 2.1.1, evaluated at $\tau = \rho^2$ and $k = \rho z$.

The framework operates on a temporal sequence of market snapshots. At each time t , a finite set of quote locations $\pi_t \subset D$ is observed together with the corresponding market implied volatilities $v_t(x)$ for $x \in \pi_t$. The GNO estimate is evaluated at these locations for calibration and empirical coverage measurement, and the constructed bands are represented over a regular grid when the complete lower and upper surfaces are projected. For a new snapshot at time t^* , the objective is to construct surfaces $v_{\text{lo}} : D \rightarrow (0, \infty)$ and $v_{\text{hi}} : D \rightarrow (0, \infty)$ around the GNO estimate, with target marginal coverage $1 - \alpha$ and the calendar and butterfly adjustments described in Section 3.5.

3.2 Nonconformity Scores

The choice of nonconformity score determines how a raw prediction error is translated into a measure of nonconformity and consequently how the resulting bands are shaped across the domain. Two scores are used. Score A normalises a price residual by the prevailing bid-ask spread, while Score B weights an implied volatility residual by the option's Vega. In both cases, the rescaling varies across the surface and reflects differences in local liquidity or price sensitivity.

3.2.1 Score A: Spread-Normalised Price Error

The first score normalises the price residual by the prevailing bid-ask spread, so that an error is judged relative to the width of the observed market quote:

$$A_t^{\text{spr}}(x) = \frac{2|r_t^P(x)|}{\max\{s_t(x), 0.10\}}, \quad r_t^P(x) = P_{t,\pm}(x, \hat{v}_t(x)) - P_{t,\pm}(x, v_t(x)). \quad (3.2)$$

This construction adapts the spread-relative pricing error used by Gonon et al. [10] for conformal calibration. Here, $P_{t,\pm}$ is the option price defined in Equation 3.1, and $s_t(x)$ is measured in the same option-price units. The floor of 0.10 avoids unstable normalisation when the recorded spread is very small or zero. When the market reference price is the bid-ask midpoint and the floor is inactive, a score of 1 corresponds to an error of half the observed spread, while a score below 1 places the model price within the quoted interval.

3.2.2 Score B: Vega-Normalised Implied Volatility Error

The second score operates directly in implied volatility space and weights the residual by the option's forward-normalised Vega. Options with above-average Vega therefore receive greater weight:

$$A_t^{\text{IV}}(x) = w_V(x; \hat{v}_t) |r_t^{\text{IV}}(x)|, \quad r_t^{\text{IV}}(x) = \hat{v}_t(x) - v_t(x), \quad (3.3)$$

where the implemented weight is

$$w_V(x; \hat{v}_t) = \max \left\{ \frac{\mathcal{V}(x, \hat{v}_t(x))}{\frac{1}{|\pi_t|} \sum_{x' \in \pi_t} \mathcal{V}(x', \hat{v}_t(x'))}, 1 \right\}. \quad (3.4)$$

The surface-normalised Vega weighting follows Gonon et al. [10]. The Vega is evaluated at the GNO estimate, and the same definition is used during calibration and band construction. Normalising by the mean Vega preserves relative sensitivity across each surface: options above the mean receive a weight greater than 1, while the floor keeps the remaining residuals on their original scale.

3.3 Rolling Window Calibration and Mondrian Binning

Financial markets are non-stationary: the distribution of prediction errors observed in one period need not remain unchanged in another. The framework therefore restricts calibration to the K most recently processed market surfaces rather than using the complete history. Let the available snapshots be ordered by time as $t_1 < t_2 < \dots$. For an evaluation at t_j , the window W_j contains up to the K most recent available snapshots preceding t_j . The window counts observed surfaces rather than elapsed calendar days, so gaps between data-collection dates do not alter its definition.

The scores from all quotes on the surfaces in W_j are pooled into

$$\mathcal{A}(W_j) = \{A_t(x) : t \in W_j, x \in \pi_t\}. \quad (3.5)$$

If $n = |\mathcal{A}(W_j)|$ and $A_{(1)} \leq \dots \leq A_{(n)}$ denote the ordered pooled scores, the implementation uses the exact conformal order statistic

$$\hat{q}_{1-\alpha} = A_{(r_n)}, \quad r_n = \min\{n, \lceil (n+1)(1-\alpha) \rceil\}. \quad (3.6)$$

Under exchangeability, this is the standard finite-sample quantile used to obtain marginal coverage at level $1 - \alpha$ [19, 2]. In the walk-forward evaluation, each new surface is evaluated using only the preceding window and is added to the window afterwards, allowing it to contribute to subsequent predictions without using future information.

A single quantile computed over the entire domain D does not reflect possible variation in the score distribution across maturities and moneyness levels. To introduce spatial adaptation, the framework applies *Mondrian conformal prediction* [20, 19]: the domain is partitioned into a regular grid of $B_\rho \times B_z$ bins, and Equation 3.6 is applied separately to the scores in each sufficiently populated bin. An evaluation point receives the quantile of the bin containing its (ρ, z) coordinates.

Because each bin contains fewer scores than the pooled calibration set, two stabilising rules are used. A bin receives its own quantile only when it contains at least 10 calibration scores; otherwise, it uses the global quantile from the same window. For a sufficiently populated bin, the raw bin quantile $q_{\text{bin}}^{\text{raw}}$ is combined with the global quantile using

$$q_{\text{bin}} = \lambda \hat{q}_{1-\alpha} + (1 - \lambda) q_{\text{bin}}^{\text{raw}}. \quad (3.7)$$

The experiments use $K = 5$ surfaces, a 4×4 partition, and $\lambda = 0.3$. This shrinkage is an empirical regularisation that reduces unstable differences between bin estimates and the global value; the coverage of the resulting spatially adaptive bands is therefore evaluated empirically. Points outside the bounds of D are assigned the global quantile.

3.4 Prediction Bands

At a new prediction time t^* , the GNO evaluates $\hat{v}_{t^*}(x)$ at the observed quote locations π_{t^*} . Each location is assigned the corresponding global or Mondrian conformal quantile. Two modes of constructing prediction bands around $\hat{v}_{t^*}(x)$ are considered, corresponding to the two nonconformity scores of Section 3.2. The resulting quote-level bounds are subsequently represented on a regular grid for the surface projection described in Section 3.5.

3.4.1 Price-Spread Mode

Under Score A, the conformal quantile is used to construct a half-width in price space, scaled by the spread at each quote:

$$h^P(x) = \frac{1}{2} q_{1-\alpha}^{\text{spr}}(x) \max\{s_{t^*}(x), 0.10\}, \quad (3.8)$$

giving price bands

$$P_{\text{lo}}(x) = P_{t^*, \pm}(x, \hat{v}_{t^*}(x)) - h^P(x), \quad P_{\text{hi}}(x) = P_{t^*, \pm}(x, \hat{v}_{t^*}(x)) + h^P(x). \quad (3.9)$$

Here, $P_{t^*, \pm}$ is the option-price function defined in Equation (3.1), using the forward and discount factor for the quote. Corresponding bands in implied volatility space are obtained by inverting the same pricing relation at the two endpoints.

3.4.2 Vega-IV Mode

Under Score B, bands are constructed directly in implied volatility space. The conformal quantile is converted to an implied volatility half-width via the same Vega weighting used in the nonconformity score:

$$h_v(x) = \frac{q_{1-\alpha}^{\text{IV}}(x)}{w_V^*(x)}, \quad w_V^*(x) = w_V(x; \hat{v}_{t^*}), \quad (3.10)$$

giving implied volatility bands

$$v_{\text{lo}}(x) = \max\{\hat{v}_{t^*}(x) - h_v(x), 10^{-4}\}, \quad v_{\text{hi}}(x) = \hat{v}_{t^*}(x) + h_v(x), \quad (3.11)$$

where 10^{-4} is the numerical floor used in the implementation to keep the lower endpoint strictly positive. If required, the corresponding price endpoints are recovered as $P_{\text{lo}}(x) = P_{t^*, \pm}(x, v_{\text{lo}}(x))$ and $P_{\text{hi}}(x) = P_{t^*, \pm}(x, v_{\text{hi}}(x))$. Both

band modes produce implied-volatility lower and upper surfaces that can be passed through the same projection described next. Their empirical comparison in Chapter 5 motivates the use of Score B as the primary construction in this thesis.

3.5 No-Arbitrage Projection

The prediction bands constructed in Section 3.4 are designed to provide statistical coverage around the GNO surface. Their construction does not directly impose the financial shape conditions introduced in Section 2.1.4. A post-processing step is therefore applied to improve the calendar and butterfly consistency of the lower and upper band surfaces. This section first summarises the corresponding conditions from the base GNO paper [10] and then describes the discrete implementation used for the conformal bands.

3.5.1 Calendar and Butterfly Constraints

Theorem 3.1 (Adapted from Theorem 2.1 of Gonon et al. [10]). *Let $\hat{v} : (0, \infty) \times \mathbb{R} \rightarrow (0, \infty)$ be continuous and satisfy:*

- (i) **Calendar arbitrage.** *For each log-moneyness $k \in \mathbb{R}$, the map $\tau \mapsto \hat{v}(\tau, k)\sqrt{\tau}$ is non-decreasing and vanishes at $\tau = 0$.*
- (ii) **Butterfly (strike) arbitrage.** *For every $\tau > 0$, the maturity slice $\hat{v}_\tau := \hat{v}(\tau, \cdot)$ is of class C^2 and satisfies*

$$\text{But}(\tau, \cdot, \hat{v}_\tau, \partial_k \hat{v}_\tau, \partial_k^2 \hat{v}_\tau) \geq 0, \quad (3.12)$$

where, for $v_0 = \hat{v}_\tau(k)$, $v_1 = \partial_k \hat{v}_\tau(k)$, $v_2 = \partial_k^2 \hat{v}_\tau(k)$,

$$\text{But}(\tau, k, v_0, v_1, v_2) = \left(1 + d_1(\tau, k, v_0) v_1 \sqrt{\tau}\right) \left(1 + d_2(\tau, k, v_0) v_1 \sqrt{\tau}\right) + v_0 v_2 \tau. \quad (3.13)$$

It also satisfies the asymptotic condition

$$\limsup_{k \uparrow \infty} \frac{\hat{v}_\tau(k)^2}{k} < \frac{2}{\tau}. \quad (3.14)$$

Then $(T, K) \mapsto BS(\tau, k, \hat{v}(\tau, k))$ defines a call price surface free of static arbitrage.

Condition (i) is the calendar spread condition already introduced in Section 2.1.4: total uncertainty should not decrease as maturity increases at the same log-moneyness. Condition (ii) is the analytic butterfly condition, for which a non-negative But functional corresponds to a non-negative risk-neutral density. The asymptotic condition concerns the behaviour of the surface outside the bounded domain used in this thesis and is therefore not part of the finite-grid post-processing step.

The base paper incorporates the calendar and butterfly conditions during GNO training through the soft penalties $\mathcal{L}_{\text{cal}}$ and $\mathcal{L}_{\text{but}}$, numbered Equations (11) and (10), respectively, in that paper. The pre-trained GNO is not retrained in this thesis. Instead, the conformal lower and upper surfaces are adjusted after band construction using discrete calendar and butterfly conditions in total implied variance. These conditions are motivated by the same financial requirements while allowing a direct and stable implementation on the transformed grid.

3.5.2 Projection Algorithm

The projection is applied independently to the lower and upper band surfaces. Each iteration uses two consecutive passes: butterfly first and calendar second. The calculations are performed in terms of *total implied variance*

$$w(\rho, z) := v(\rho, z)^2 \rho^2, \quad \tau = \rho^2, \quad k = \rho z. \quad (3.15)$$

On the transformed grid, the calendar pass requires total implied variance to be non-decreasing between consecutive maturity coordinates at fixed z :

$$w(\rho_i, z) \leq w(\rho_{i+1}, z), \quad \rho_i < \rho_{i+1}. \quad (3.16)$$

The butterfly pass applies a chord condition across z at each fixed maturity. For three consecutive grid points $z_1 < z_2 < z_3$,

$$w(\rho, z_2) \leq \frac{\Delta z_2 w(\rho, z_1) + \Delta z_1 w(\rho, z_3)}{\Delta z_1 + \Delta z_2}, \quad \Delta z_1 = z_2 - z_1, \quad \Delta z_2 = z_3 - z_2. \quad (3.17)$$

At a fixed ρ , the transformation $k = \rho z$ is linear, so the ordering of points within a maturity slice is preserved. Equations 3.16 and 3.17 are the calendar and butterfly conditions actually enforced by the post-processing routine.

The observed option quotes lie at irregular (ρ, z) locations. For each evaluation surface, their band values are mapped by linear interpolation to a rectangular grid formed from the observed distinct ρ values, rounded to four decimal places, and 50 equally spaced z values. Nearest-neighbour interpolation fills grid locations outside the convex hull. The first pass then applies Equation 3.17 to each maturity slice. If a middle point is above its chord value, its total variance is moved part-way towards that value:

$$w(\rho, z_2) \leftarrow w(\rho, z_2) + \eta(w_{\text{chord}} - w(\rho, z_2)), \quad w_{\text{chord}} = \frac{\Delta z_2 w(\rho, z_1) + \Delta z_1 w(\rho, z_3)}{\Delta z_1 + \Delta z_2}. \quad (3.18)$$

The implementation uses $\eta = 0.8$. The second pass applies Equation 3.16 down each fixed- z column. If $w(\rho_i, z) > w(\rho_{i+1}, z) + \varepsilon$, the two values are replaced by their average. After each pass, the corrected values are converted back to implied volatility using $v = \sqrt{w/\rho^2}$. The passes are repeated until the change between iterations is below the tolerance $\varepsilon = 10^{-6}$ or 500 iterations have been completed. This is summarised in Algorithm 1.

Algorithm 1 Conformal bands with no-arbitrage projection

Require: Trained GNO $\hat{F}^\theta$, rolling window W , spatial binning of D , tolerance ε , coverage level $1 - \alpha$

- 1: **Calibration:** compute nonconformity scores (Equation 3.2 and/or Equation 3.3) in each bin; store the empirical quantile $\hat{q}_{1-\alpha}$ per Equation 3.6
 - 2: **Predict:** evaluate $\hat{v} = \hat{F}^\theta$ on the observed quotes; compute $v_{\text{lo}}, v_{\text{hi}}$ via the price-spread or Vega-IV mode
 - 3: **Interpolate:** map both IV bands to the rectangular (ρ, z) grid
 - 4: **for** each of v_{lo} and v_{hi} **do**
 - 5: apply the butterfly pass across z at each maturity
 - 6: apply the calendar pass across maturity at fixed z
 - 7: repeat until convergence or the iteration limit
 - 8: **end for**
 - 9: **Map back:** interpolate both projected surfaces to the observed quote locations
 - 10: **Check:** count remaining calendar and butterfly violations and lower–upper crossings
 - 11: **return** projected IV bands $(\tilde{v}_{\text{lo}}, \tilde{v}_{\text{hi}})$ and projection diagnostics
-

The projected grid values are mapped back to the observed quote locations by linear interpolation; if a mapped value is unavailable, the original band endpoint at that quote is retained. Because the lower and upper surfaces are projected independently, lower–upper crossings are checked both on the grid and after mapping back. Corresponding price endpoints can, when required, be recovered using the option-price function $P_{t^*, \pm}$ defined in Equation (3.1).

3.5.3 Coverage after Projection

The direct conformal construction provides the bands before projection. Since the projection can adjust their endpoints, coverage after projection is evaluated empirically rather than inferred from the pre-projection statement. The evaluation therefore reports matched coverage before and after projection on the same test quotes, together with the remaining calendar and butterfly counts and the number of lower–upper crossings.

3.6 Evaluation Metrics

The framework is evaluated in walk-forward order. Each evaluation surface is assessed using only the preceding surfaces in the rolling calibration window and is added to the window afterwards. Consequently, an evaluation surface is never used to calibrate its own prediction bands.

Coverage: For an evaluation surface at t^* , within-surface coverage is the fraction of observed market implied volatilities contained in the band:

$$\widehat{\text{Cov}}_{t^*} = \frac{1}{|\pi_{t^*}|} \sum_{x \in \pi_{t^*}} \mathbf{1}\{v_{t^*}(x) \in [v_{\text{lo}}(x), v_{\text{hi}}(x)]\}. \quad (3.19)$$

It is compared with the nominal target $1 - \alpha$. Coverage is computed separately for the global and Mondrian quantiles and, where relevant, before and after projection. The per-surface values are then summarised across the evaluation period.

Sharpness: The IV band width is $v_{\text{hi}}(x) - v_{\text{lo}}(x)$. Its mean and median are calculated over the observed quote locations of each evaluation surface after the projected grid has been mapped back to those locations. Narrower bands are preferred when coverage is comparable.

Calibration diagnostics: The global quantile and the mean, minimum, and maximum Mondrian quantiles are recorded as the rolling window advances. The number of calibration scores and the mean score per surface are also retained. These quantities show how the calibration threshold changes through time and across regions of the (ρ, z) domain.

Calendar and butterfly diagnostics: The calendar and butterfly violations defined by the grid-based checks of Section 3.5.2 are counted before and after projection for the lower and upper band surfaces. The evaluation also records whether the projection converged and counts lower–upper crossings on the grid and at the observed quote locations.

Point-prediction error: The mean absolute error between the GNO prediction $\hat{v}_{t^*}(x)$ and the market implied volatility $v_{t^*}(x)$ is recorded as a common reference value. Because both score modes use the same trained GNO predictions, differences in their coverage and sharpness arise from the nonconformity score and band construction rather than from different point predictions.

3.7 Summary

This chapter has presented the methodology for conformal uncertainty quantification around GNO-based implied volatility surfaces. The framework combines two economically motivated nonconformity scores, walk-forward rolling calibration, Mondrian spatial binning, prediction bands in price and implied-volatility space, and the calendar and butterfly projection of the lower and upper band surfaces. Its evaluation uses matched coverage, sharpness, calibration, projection, crossing, and point-prediction diagnostics. These components provide the methodological basis for assessing the six research objectives set out in Chapter 1. The following chapter describes their software implementation, and Chapter 5 reports the empirical comparison of Score A and Score B on the S&P 500 options data.

Chapter 4

Implementation

4.1 Software Architecture

The conformal uncertainty quantification framework is implemented as a self-contained Python package, `volatility_smoothing/conformal/`, that operates entirely as a post-hoc wrapper around the pre-trained GNO. No modification is made to the base model's architecture, weights, or inference code; the package receives the GNO's point predictions as inputs and returns calibrated prediction bands as outputs. The package is structured into five modules, each encapsulating a component of the pipeline described in Chapter 3. Table 4.1 lists each module alongside its primary interface and its role in the framework.

Module	Primary interface	Role
<code>nonconformity_scores.py</code>	<code>SpreadNormalizedScore</code> <code>VegaNormalizedScore</code>	Score A: spread-normalised price error Score B: Vega-normalised IV error
<code>rolling_window.py</code>	<code>RollingWindowCalibration</code>	FIFO window of the K most recent calibration surfaces
<code>mondrian.py</code>	<code>MondrianConformal</code>	Spatial binning and per-bin quantile computation
<code>prediction_bands.py</code>	<code>PredictionBands</code>	Band construction in price-spread and Vega-IV modes
<code>no_arbitrage_projection.py</code>	<code>project_to_arbitrage_free</code>	Alternating butterfly and calendar projection

Table 4.1: Modules of the `volatility_smoothing/conformal/` package and their roles.

The five modules are wired together in the experiment notebook `conformal_prediction.ipynb`, which serves as the authoritative integration point for the reported pipeline and is described in Section 4.8. The notebook also contains a small verified helper layer for the exact conformal order statistic, consistent Score B weights, and projection diagnostics. These helpers retain the structure of the package while ensuring that the final experiment follows the methodology of Chapter 3. Each component is described in turn in the sections that follow.

4.2 Nonconformity Score Computation

Both nonconformity scores are implemented in `nonconformity_scores.py` as subclasses of a common `NonconformityScore` base class, which defines a single `compute` method accepting the GNO's predicted implied volatilities, the market implied volatilities, and a dictionary of auxiliary data. A top-level dispatcher function, `compute_scores_from_data`, selects the appropriate scorer based on a `score_type` argument and assembles the auxiliary data from the raw surface dictionary produced by the dataloader before delegating to the scorer.

Score A (`SpreadNormalizedScore`) works in price space. It takes three steps: (1) compute predicted and market

prices from the corresponding implied volatilities using the Black-Scholes formula in transformed coordinates - $d_1 = -z/v + v\rho/2$ and $d_2 = -z/v - v\rho/2$, with call and put prices handled separately based on the sign of the log-moneyness $k = \rho z$, (2) read bid and ask prices directly from the dataloader as dollar-denominated quantities and floor the spread at 0.10, (3) divide twice the absolute price residual by the floored spread, per Equation 3.2. When the market-price reference is the bid-ask midpoint and the observed spread exceeds the floor, a score below 1 indicates that the model price lies within the observed bid-ask interval. More generally, the score measures the price residual relative to the floored spread.

Score B (`VegaNormalizedScore`) works directly in implied volatility space. In the final experiment, the notebook helper `score_from_data_corrected` computes forward-normalised Vega at the GNO-predicted implied volatility, divides it by the mean Vega across the surface, and multiplies the absolute IV residual by the resulting relative weight. The completed weight ratio $\mathcal{V}/\bar{\mathcal{V}}$ is floored at `min_weight= 1.0`, so options with above-average Vega are up-weighted and no option is weighted below 1. Applying the floor to the completed ratio avoids collapsing the relative weights when forward-normalised Vega values are below 1. The same weighting convention is used when the Score B prediction bands are constructed.

4.3 Rolling Window Calibration

The rolling window is implemented in `rolling_window.py` as a `RollingWindowCalibration` class that maintains an ordered list of `CalibrationSurface` dataclasses, each storing the (ρ, z) coordinates and nonconformity scores for one market snapshot. The window operates as a FIFO buffer of fixed size K : `add_surface` appends a new surface and, if the list exceeds K entries, discards the oldest. `get_calibration_data` concatenates the coordinates and scores across all surfaces currently in the window into flat arrays, which are passed directly to `MondrianConformal` for quantile computation. The window makes no assumption about the time spacing between surfaces - it counts snapshots, not elapsed time.

Two state queries are available: `is_ready` returns true as soon as at least one surface has been added, and `is_full` returns true once the window has reached capacity K . In the experiment pipeline, quantile computation begins immediately after the first surface is added, so the Mondrian calibration runs on a growing window during the early calibration surfaces and on a full window of K surfaces thereafter.

4.4 Mondrian Spatial Binning

Mondrian binning is implemented in `mondrian.py` as a `MondrianConformal` class. At initialisation, the domain $D = (0.01, 1) \times (-1.5, 0.5)$ is partitioned into a regular $B_\rho \times B_z$ grid of `MondrianBin` objects using evenly spaced edges along each axis. Each bin stores the list of nonconformity scores from the rolling window that fall within its boundaries.

Calibration proceeds in two passes. The final experiment uses `ExactOrderStatisticMondrian`, a subclass defined in `conformal_prediction.ipynb`, to apply the finite-sample conformal order statistic in Equation 3.6. First, the global quantile is computed over all scores in the window as a fallback. Second, per-bin quantiles are computed for bins that contain at least `min_samples_per_bin = 10` scores; bins below this threshold are assigned the global quantile directly. For bins that meet the threshold, the raw bin quantile is shrunk towards the global quantile by a factor λ :

$$q_{\text{bin}} = \lambda \hat{q}_{1-\alpha} + (1 - \lambda) q_{\text{bin}}^{\text{raw}}, \quad (4.1)$$

with $\lambda = 0.3$ in the experiments of Chapter 5. This prevents bins that marginally exceed the minimum count from producing extreme quantile estimates while preserving the spatial adaptivity of well-populated bins.

At test time, `get_quantiles` assigns each test point the quantile of the bin containing its (ρ, z) coordinates. Points that fall outside the domain boundaries are assigned the global quantile. Bin lookup uses `np.searchsorted` on the precomputed bin edges for efficient assignment.

4.5 Prediction Band Construction

Band construction is provided by `prediction_bands.py` through the `PredictionBands` class. The construction accepts the GNO point predictions, the per-point global or Mondrian conformal quantiles, and the market data required by the chosen mode, and returns lower and upper bounds in both IV and price space. In the final pipeline, Score A uses `PredictionBands.compute`, while Score B uses the notebook helper `vega_iv_bands_corrected` so that scoring and band construction share the same forward-normalised Vega weights.

Under the Vega-IV mode, the corrected helper takes four steps: (1) compute forward-normalised Vega at each quote using the GNO-predicted IV, (2) form the normalised weight $w_V = \mathcal{V}/\bar{\mathcal{V}}$ and floor it at `min_vega= 1.0`, (3) divide the per-point conformal quantile by the weight to obtain the IV half-width $h_v = q/w_V$, and (4) subtract and add h_v from the GNO prediction to obtain v_{lo} and v_{hi} , with v_{lo} clipped at 10^{-4} to ensure a strictly positive lower bound. Price bands are then recovered by evaluating the Black-Scholes formula at v_{lo} and v_{hi} .

Under the price-spread mode, `compute` takes three steps: (1) compute the GNO predicted price using the Black-Scholes formula, (2) read bid and ask prices from the dataloader and floor the spread at `min_spread= 0.10`, (3) add and subtract the price half-width $h_P = (q/2) \cdot s$ from the predicted price to get P_{lo} and P_{hi} . IV bands are then obtained by inverting the Black-Scholes price at each endpoint via `implied_volatility_from_price`, which root-finds per point using Brent's method over the interval $[10^{-4}, 5.0]$. Points where no sign change is found within this interval are clipped to the nearest bound rather than raising an error.

One important distinction is that the `black_scholes_price` function in this module produces dollar-denominated prices - that is, $\text{discount} \times \text{forward} \times \text{BS}(\tau, k, v)$ - rather than the forward-normalised, dimensionless $\text{BS}_{\pm}(x, v)$ used in Chapter 3. These currency units are used consistently with the observed bid and ask prices in Score A. Score B is constructed directly in implied-volatility space using the forward-normalised Vega weights described above.

4.6 No-Arbitrage Projection

The no-arbitrage projection is implemented in `no_arbitrage_projection.py` and consists of three functions: `check_butterfly_arbitrage`, `check_calendar_arbitrage`, and `project_to_arbitrage_free`. In the final notebook, the scattered (ρ, z) band values are represented on a rectangular grid whose rows correspond to the available ρ values, with maturities $\tau = \rho^2$, and whose columns contain 50 evenly spaced z values. The mapping is performed with `scipy.interpolate.griddata` using linear interpolation, with nearest-neighbour interpolation as a fallback outside the convex hull of the scattered data.

The two check functions each return an `ArbitrageConstraints` dataclass recording the violation count and maximum violation magnitude. `check_butterfly_arbitrage` iterates over each maturity slice, computes total variance $w = v^2\tau$ across the z direction, and flags any middle grid point where w exceeds the chord value between its two neighbours by more than a tolerance $\varepsilon = 10^{-6}$. `check_calendar_arbitrage` iterates over each fixed- z column and flags any consecutive maturity pair where w decreases.

`project_to_arbitrage_free` runs an alternating projection loop. Each iteration performs two passes in order: (1) the butterfly pass iterates over each maturity slice and, for each violating middle grid point, moves w part-way towards the chord value by a step size $\eta = 0.8$; (2) the calendar pass iterates over each fixed- z column and, for each violating consecutive pair, replaces both values with their average. After each pass, total variance is converted back to implied volatility via $v = \sqrt{w/\tau}$. The loop terminates when the L2 norm of the surface change between successive iterations falls below ε , or after a maximum of 500 iterations. If the input surface already satisfies both discrete checks, the function returns immediately without iterating.

The lower and upper band surfaces are projected independently. Their corrected grid values are then mapped back to the original scattered (ρ, z) coordinates via a second `scipy.interpolate.griddata` call, with any points outside the grid convex hull filled from the corresponding pre-projection band values. The resulting scattered values form the final $\tilde{v}_{lo}$ and $\tilde{v}_{hi}$ used for coverage and band-width evaluation in Chapter 5. Lower-upper crossings are counted after projection as an additional consistency check.

4.7 Data Acquisition and Preprocessing

The SPX option chains used in the empirical study were collected through the OpenBB Python interface using its Cboe provider and the symbol SPX [15]. A scheduled retrieval script requested the available option chain and saved the returned expiration dates, strikes, option types, bid prices, and ask prices as CSV files. The files were written in the column format expected by the existing WRDSOptionsDataset loader. The name of this loader therefore refers to the compatible file schema and does not imply that the observations were obtained from WRDS.

During this transformation, the calendar date on which the retrieval completed was stored in the date field. The dates used to order the surfaces in the notebook and figures are consequently collection dates rather than separate exchange-provided quote timestamps. The `am_settlement` field was set to one as a compatibility field required by the loader’s filtering step; it is not used in this thesis as an empirical classification of the settlement convention of each individual contract. After loading, the existing preprocessing pipeline computes the time to maturity, forward price, discount factor, log-moneyness, and implied volatility and restricts the observations to the transformed GNO domain.

4.8 Experiment Notebook

The experiment notebook `conformal_prediction.ipynb` is the authoritative integration point for the reported pipeline. It loads the pre-trained GNO checkpoint and the WRDS-compatible OpenBB files via `WRDSOptionsDataset` and `GNOOptionsDataset`, and splits the available surfaces into a calibration set of 15 surfaces and a test set of 41 surfaces. The key hyperparameters used throughout the experiments are listed in Table 4.2.

Hyperparameter	Symbol	Value
Coverage level	$1 - \alpha$	90%
Rolling window size	K	5 surfaces
Mondrian bins	$B_\rho \times B_z$	4×4
Shrinkage factor	λ	0.3
Min samples per bin		10
Projection step size	η	0.8
Max projection iterations		500

Table 4.2: Hyperparameters used in all experiments.

The pipeline is encapsulated in a `run_conformal_pipeline` function that accepts a `score_type` argument (`‘vega’` for Score B or `‘spread’` for Score A) and runs the full calibration and test loop for that score. During calibration, the function iterates over the 15 calibration surfaces in order: for each surface, it runs the GNO forward pass, computes nonconformity scores, adds them to the rolling window, and immediately recalibrates the global and Mondrian quantiles on the updated window. During testing, it iterates over the 41 test surfaces: for each, it runs the GNO forward pass, retrieves both global and per-point Mondrian quantiles, constructs both variants of the prediction bands, applies the no-arbitrage projection inline, and records coverage, band width, calendar and butterfly violation counts before and after projection, lower–upper crossings, and point-prediction errors. After each test surface is evaluated, its scores are added to the rolling window so that the calibration set continues to slide forward through time. The function is called once for each score, and the results are compared in Chapter 5.

Figure 4.1 illustrates the structure of the dataset used throughout all experiments. The top panel shows the number of option quotes per surface across all 56 successfully collected surfaces, with calibration surfaces (indices 0–14) shaded in blue and test surfaces (indices 15–55) in red; the bottom panel displays the timeline of collection dates with the rolling window annotation. The collection dates span 11 December 2025 to 15 August 2026, and each processed surface contains between 8,677 and 11,325 quotes, with a mean of approximately 10,070. The calibration set covers collection dates from 11 December 2025 to 23 January 2026, and the test set covers collection dates from 29 January 2026 to 15 August 2026.

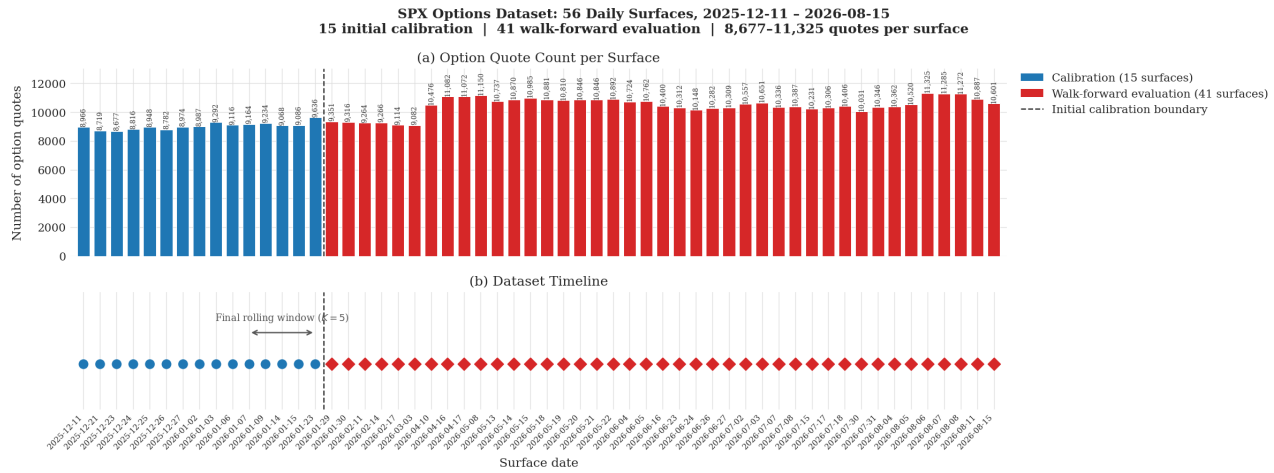


Figure 4.1: SPX options dataset: option quote count per surface (top) and collection timeline (bottom) across 56 successfully collected surfaces from December 2025 to August 2026. Blue indicates calibration surfaces (indices 0–14); red indicates test surfaces (indices 15–55).

Chapter 5

Experiments and Results

This chapter reports the empirical evaluation of the conformal uncertainty quantification framework developed in Chapter 3 and implemented in Chapter 4. The evaluation is organised around the six research objectives set out in Section 1.2: whether the framework achieves the nominal coverage level on held-out market data, whether the rolling window and Mondrian spatial binning produce stable and spatially adaptive quantiles, whether the no-arbitrage projection reliably eliminates butterfly and calendar violations, and whether the resulting bands are sharp enough to be practically informative. Section 5.1 specifies the dataset and evaluation protocol. Section 5.2 examines quantile behaviour during the calibration phase. Section 5.3 presents the no-arbitrage projection results for both scores and establishes why Score B is the recommended construction. Section 5.4 reports the full Score A versus Score B comparison across coverage, sharpness, and projection tractability. Sections 5.5 and 5.6 then report the detailed coverage and sharpness results for Score B as the primary framework. Section 5.7 summarises the findings.

5.1 Experimental Setup

The empirical dataset comprises 56 S&P 500 (SPX) option surface snapshots successfully collected between 11 December 2025 and 15 August 2026. The option chains were obtained through OpenBB using its Cboe provider and stored in a WRDS-compatible CSV schema, as described in Section 4.7. They are loaded by the `WRDSOptionsDataset` class, whose name refers to the expected schema rather than the source of the observations. The first 15 surfaces, with collection dates from 11 December 2025 to 23 January 2026, form the initial calibration set. The remaining 41 surfaces, with collection dates from 29 January 2026 to 15 August 2026, are evaluated sequentially using a walk-forward procedure. The collection dates are irregularly spaced because provider-side API errors and timeouts prevented successful retrieval on some scheduled dates. The analysis uses all successfully obtained surfaces in chronological order so that no later surface enters an earlier calibration window. Each processed surface contains between 8,677 and 11,325 option quotes in the transformed coordinate domain $D = (0.01, 1) \times (-1.5, 0.5)$.

The structure of the dataset is illustrated in Figure 4.1 of Chapter 4, which shows the option quote count per surface and the calibration/evaluation timeline across all 56 surfaces.

The point predictor underlying the framework is the pre-trained GNO of Gonon, Jacquier, and Wiedemann [10], loaded from the checkpoint `train/store/9448705/checkpoints/checkpoint_final.pt`. The model has 102,529 parameters and is run in inference mode without any retraining or fine-tuning on the SPX data used here. The conformal layer is applied post-hoc, as described in Chapter 3: the GNO point predictions are fixed, and all calibration and band construction is performed solely through the conformal pipeline. All hyperparameters are as listed in Table 4.2 of Chapter 4: coverage level $1 - \alpha = 90\%$, rolling window $K = 5$ surfaces, 4×4 Mondrian binning, shrinkage $\lambda = 0.3$, minimum bin population 10, projection step size $\eta = 0.8$, and maximum 500 projection iterations.

The primary evaluation is conducted using Score B (Vega-normalised implied volatility error, Section 3.2.2) with Vega-IV bands (Section 3.4.2), which is the construction recommended by this thesis. Score A (spread-normalised price error, Section 3.2.1) with price-spread bands (Section 3.4.1) is evaluated under identical conditions as an

ablation, to examine how the choice of nonconformity score affects coverage stability, sharpness, and the tractability of the no-arbitrage projection. Because both scores are applied to the same GNO predictions on the same dataset, any differences in the reported metrics are attributable solely to the nonconformity score and band construction.

5.2 Calibration Behaviour

During the calibration phase the pipeline processes each of the 15 calibration surfaces in chronological order. After each surface is added to the rolling window, the global and Mondrian quantiles are recomputed on the scores currently in the window. This section reports the behaviour of those quantiles for Score B across the calibration period, since Score B is the primary construction and its quantiles are the ones used for the main coverage evaluation in Section 5.5.

5.2.1 Rolling Window Warm-up

The rolling window accumulates surfaces until it reaches its capacity of $K = 5$. During this warm-up period, surfaces 0 through 4 (11 December 2025 to 25 December 2025), the calibration set grows with each step. The global 90% conformal quantile is 0.016228 after the first surface and reaches 0.015954 once the window is full at surface 4. These values are closely grouped: each surface in this period contributes between 8,677 and 8,966 quotes to the window, so even the one-surface window contains a large number of calibration points and produces a stable quantile estimate.

From surface 5 onward the window operates in steady state, discarding the oldest surface as each new one is added. The global quantile fluctuates as the window advances but generally increases across the calibration period, reaching 0.017303 at surface 10 (7 January 2026) and 0.017689 at surface 14 (23 January 2026), the final calibration surface. This behaviour shows how the rolling procedure adapts the calibration threshold as the recent score distribution changes.

5.2.2 Global versus Mondrian Quantiles

Throughout the calibration period the Mondrian quantiles span a substantially wider range than the single global quantile, confirming that Score B nonconformity varies across the implied volatility surface. At surface 0, when the window contains only a single surface, the Mondrian bin quantiles range from 0.012505 to 0.033761, a ratio of approximately 2.7 between the lowest and highest bin. By surface 14, after the window has operated in steady state for ten surfaces, the range is $[0.011950, 0.040613]$, a ratio of approximately 3.4.

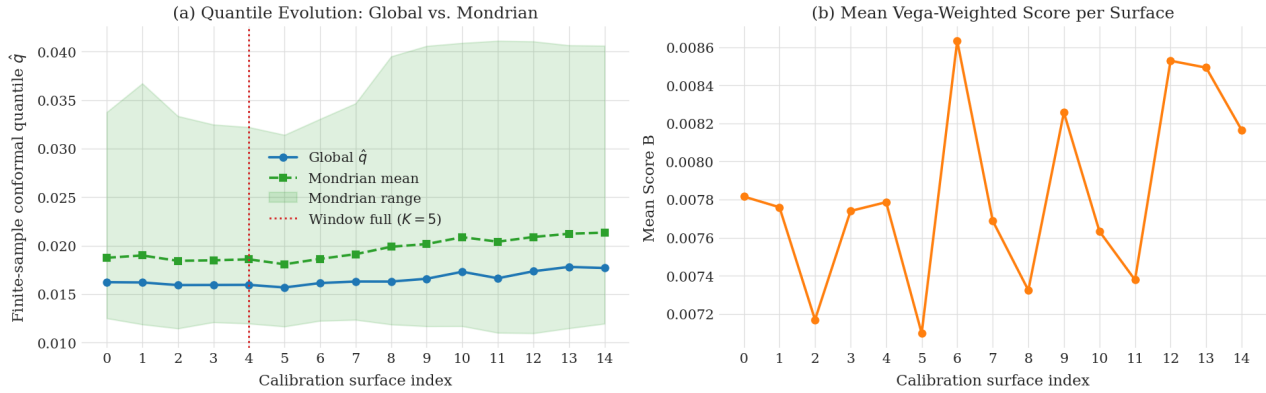
This spatial variation is the type of heterogeneity that Mondrian conformal prediction [19, 2] is designed to capture. The global construction applies one calibration threshold across the full domain, whereas the Mondrian construction assigns each point the quantile of the bin containing its (ρ, z) coordinates. Both constructions subsequently use the local Vega weight when converting the quantile into an implied volatility band. The practical effect of the bin-specific thresholds on evaluation coverage and sharpness is examined in Sections 5.5 and 5.6.

Table 5.1 reports the global and Mondrian quantile range at each of the 15 calibration surfaces for Score B, and Figure 5.1 displays the same information graphically alongside the per-surface mean Score B value.

Idx	Date	Global $\hat{q}$	Mondrian min	Mondrian max
0	2025-12-11	0.016228	0.012505	0.033761
1	2025-12-21	0.016201	0.011879	0.036719
2	2025-12-23	0.015930	0.011457	0.033356
3	2025-12-24	0.015941	0.012101	0.032480
4	2025-12-25	0.015954	0.011960	0.032213
5	2025-12-26	0.015683	0.011660	0.031414
6	2025-12-27	0.016138	0.012247	0.033058
7	2026-01-02	0.016299	0.012350	0.034665
8	2026-01-03	0.016293	0.011870	0.039508
9	2026-01-06	0.016585	0.011685	0.040585
10	2026-01-07	0.017303	0.011696	0.040894
11	2026-01-09	0.016635	0.011013	0.041116
12	2026-01-14	0.017353	0.010963	0.041055
13	2026-01-15	0.017806	0.011478	0.040656
14	2026-01-23	0.017689	0.011950	0.040613

Table 5.1: Score B calibration quantiles across 15 surfaces.

Score B Calibration Behaviour (15 surfaces, Dec 2025 - Jan 2026)

Figure 5.1: Score B global quantile $\hat{q}$, Mondrian quantile range, and per-surface mean score across the 15 calibration surfaces.

5.3 No-Arbitrage Projection

The conformal construction is complemented by a post-processing step that improves the financial consistency of the lower and upper band surfaces. As discussed in Section 3.5.2, the routine operates on a rectangular (ρ, z) grid and alternates between calendar and butterfly conditions. The calendar step enforces non-decreasing total variance across maturities at fixed z , while the butterfly step enforces convexity of total implied variance across z for each maturity. The projection is applied to both band surfaces before the final coverage and violation counts are evaluated.

5.3.1 Aggregate Violation Counts

For each evaluation surface, calendar and butterfly violations are counted separately on the lower and upper Mondrian bands and then summed. Table 5.2 reports the aggregate counts over all 41 walk-forward evaluation surfaces. It also

reports quote-level band crossings after projection, defined as locations where the projected lower bound exceeds the projected upper bound.

Score	Before	After	Reduction	Crossings after
Score A (spread)	82,190	43,471	47.1%	0
Score B (Vega)	59,414	0	100.0%	0

Table 5.2: Combined calendar and butterfly violation counts on the lower and upper Mondrian bands across 41 walk-forward evaluation surfaces.

Figures 5.2 and 5.3 illustrate the calendar and butterfly conditions on the lower Score B band for the first evaluation surface (surface 15, 29 January 2026). The left panels show selected slices before projection and the right panels show the corresponding slices after projection. Because the figures display selected lower-band slices, their violation counts differ from the aggregate totals in Table 5.2.

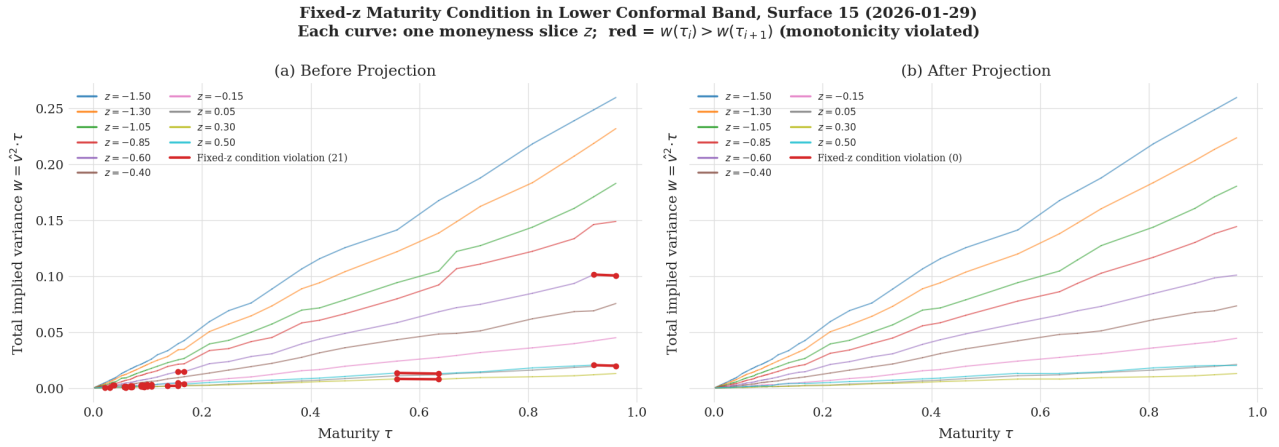


Figure 5.2: Calendar condition in selected slices of the lower Score B band for surface 15, before and after projection.

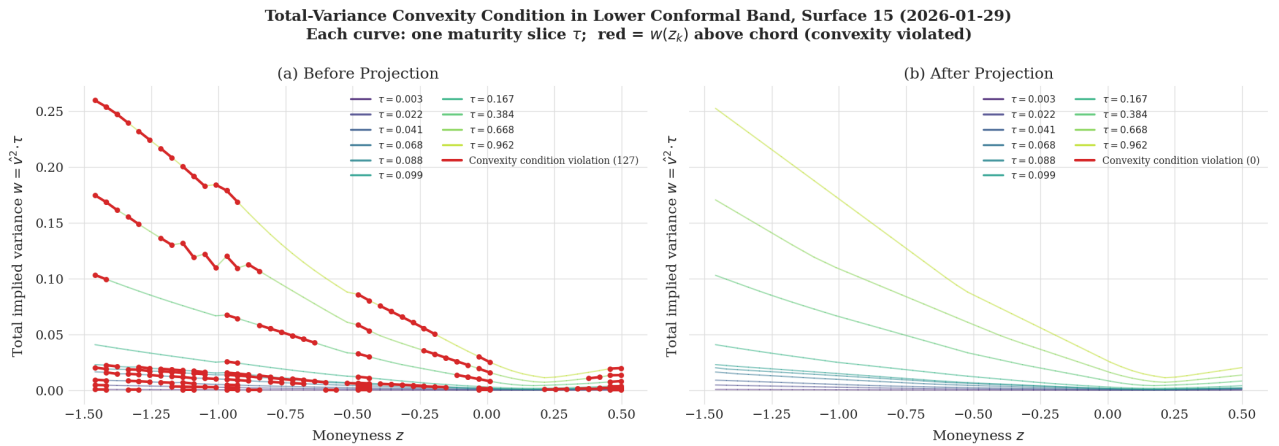


Figure 5.3: Butterfly condition in selected maturity slices of the lower Score B band for surface 15, before and after projection.

5.3.2 Score B: Elimination of Violations

For Score B, the projection reaches zero residual calendar and butterfly violations on every evaluation surface, reducing the combined lower-and-upper count from 59,414 to zero. The projected bands also have zero quote-level crossings on all 41 surfaces. This confirms that the projection consistently enforces the calendar and butterfly conditions used by the grid-based routine across all evaluation surfaces.

5.3.3 Score A: Partial Violation Reduction

For Score A, the combined violation count decreases from 82,190 to 43,471, a reduction of 47.1%, but does not reach zero on any of the 41 evaluation surfaces. The projected Score A bands nevertheless retain the correct lower–upper ordering, with zero quote-level crossings. Together with the much greater band width reported in Section 5.4.2, the remaining violations make Score A less suitable than Score B for the final framework. Score A is therefore retained as a comparative baseline rather than used as the primary band construction.

5.4 Nonconformity Score Comparison

The projection results in Section 5.3 show different behaviour for the two scores: Score B reaches zero residual calendar and butterfly violations under the grid-based routine, whereas Score A substantially reduces the counts but retains residual violations. This section compares score magnitude, band width, and coverage stability to explain why Score B is selected as the primary construction.

5.4.1 Score Magnitude and Quantile Scale

The two scores use different normalisations and therefore operate at different numerical scales. Score B is a Vega-weighted IV residual. Because both experiments use the same GNO predictions, their mean prediction error is identical at $MAE = 0.006789$ IV units. For Score B, the global calibration quantile is 0.017689 on the final calibration surface. Score A is a spread-normalised price residual. On the first calibration surface (11 December 2025), the median Score A value is 0.9717 and the 90th percentile is 3.6784, with the maximum reaching 33.0850. Its global conformal quantile is 3.683856 on that surface and increases to values of approximately 20–22 as the rolling window advances. Score A divides the price residual by the bid–ask spread, subject to the implemented minimum spread of 0.10. Because the Score A and Score B quantiles enter different band-construction formulas, Equations 3.8 and 3.10, their raw magnitudes are not directly comparable. Their practical effects are instead compared below using IV-space band width and coverage.

5.4.2 Band Width

Table 5.3 reports the mean and median IV band widths for both scores across the 41 evaluation surfaces, after Mondrian quantile assignment and no-arbitrage projection. Each entry is the average of the corresponding per-surface statistic.

Metric	Score A (spread)	Score B (vega)
Mean band width (IV)	0.1491	0.0273
Median band width (IV)	0.1148	0.0262

Table 5.3: Average per-surface mean and median IV band widths across the 41 evaluation surfaces.

Score A produces bands that are on average 5.5 times wider than Score B in mean IV width and 4.4 times wider in median IV width. The Score A bands are therefore considerably less sharp. Score B achieves a mean width of

0.027 IV units while retaining mean coverage above the nominal target, as reported below. Its local widths reflect the bin-specific Mondrian quantiles together with the subsequent projection.

5.4.3 Coverage Stability

Table 5.4 reports the post-projection per-surface Mondrian coverage for both scores alongside the overall mean and standard deviation across all 41 evaluation surfaces.

Idx	Date	Score A coverage	Score B coverage
15	2026-01-29	99.43%	91.44%
16	2026-01-30	89.60%	92.28%
17	2026-02-11	91.77%	91.91%
18	2026-02-14	93.76%	91.96%
19	2026-02-17	96.61%	91.52%
20	2026-03-03	96.14%	92.15%
21	2026-04-10	90.43%	92.03%
22	2026-04-16	95.78%	93.40%
23	2026-04-17	95.48%	92.23%
24	2026-05-08	97.33%	95.60%
25	2026-05-13	94.67%	90.98%
26	2026-05-14	97.53%	91.72%
27	2026-05-15	99.05%	94.88%
28	2026-05-18	94.54%	90.75%
29	2026-05-19	97.22%	91.66%
30	2026-05-20	98.16%	92.72%
31	2026-05-21	98.40%	91.33%
32	2026-05-22	97.19%	93.89%
33	2026-06-04	93.72%	89.92%
34	2026-06-05	85.65%	90.78%
35	2026-06-16	89.42%	88.35%
36	2026-06-23	93.72%	92.62%
37	2026-06-24	95.43%	90.68%
38	2026-06-26	97.18%	94.76%
39	2026-06-27	99.20%	89.60%
40	2026-07-02	93.81%	91.80%
41	2026-07-03	95.16%	87.57%
42	2026-07-07	92.02%	90.77%
43	2026-07-08	94.65%	90.64%
44	2026-07-15	96.64%	94.45%
45	2026-07-17	97.42%	89.13%
46	2026-07-18	96.37%	90.92%
47	2026-07-30	98.03%	89.87%
48	2026-07-31	96.08%	89.72%
49	2026-08-04	93.06%	90.41%
50	2026-08-05	98.69%	91.19%
51	2026-08-06	92.28%	88.66%
52	2026-08-07	86.73%	91.51%
53	2026-08-08	90.81%	92.37%
54	2026-08-11	93.35%	87.56%
55	2026-08-15	92.77%	89.07%
Mean		94.76%	91.34%
Std		3.31%	1.86%

Table 5.4: Post-projection per-surface Mondrian coverage for Score A and Score B across all 41 evaluation surfaces. Target: $1 - \alpha = 90\%$.

Figure 5.4 shows the post-projection per-surface coverage for both scores, together with the 90% nominal target. Figure 5.5 separately shows their mean IV band widths across the same 41 evaluation surfaces.

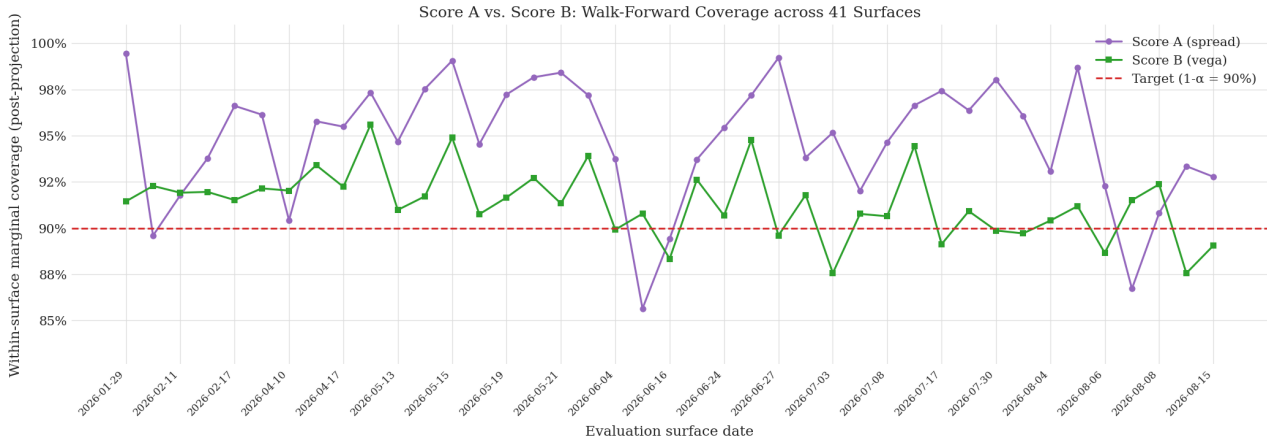


Figure 5.4: Post-projection per-surface Mondrian coverage for Score A and Score B across the 41 evaluation surfaces.

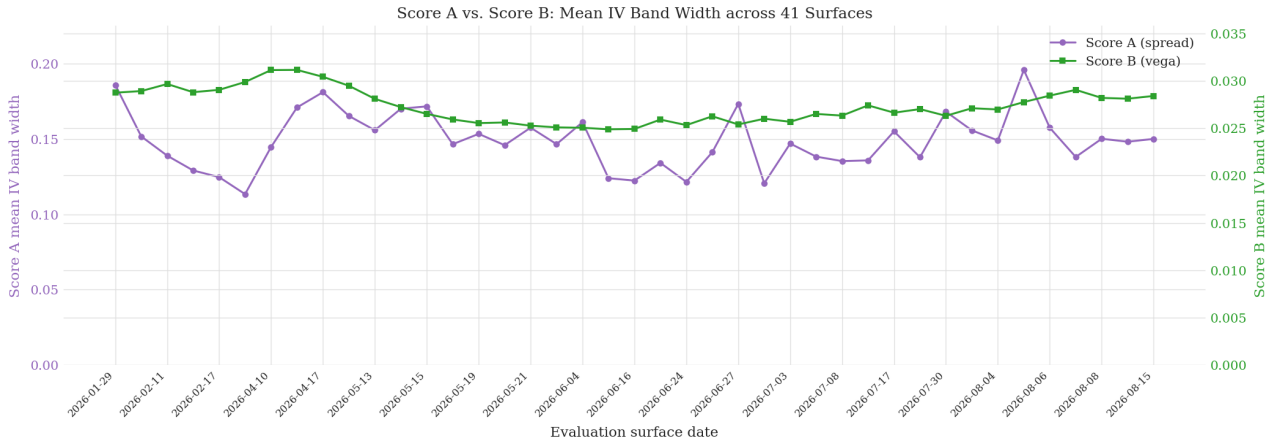


Figure 5.5: Post-projection mean IV band width for Score A and Score B across the 41 evaluation surfaces. The two vertical axes show the different width scales of the two scores.

Both scores achieve mean coverage above the 90% nominal target over the full evaluation period. However, their coverage distributions differ. Score A's standard deviation of 3.31 percentage points is 1.8 times larger than Score B's 1.86 percentage points. Score A's coverage ranges from 85.65% (surface 34) to 99.43% (surface 15), a spread of nearly 14 percentage points, showing greater variation across evaluation surfaces.

Score B's coverage ranges from 87.56% to 95.60%, a spread of approximately 8.0 percentage points. It falls below the 90% target on 10 of the 41 evaluation surfaces, whereas Score A falls below the target on 4 surfaces. Although Score A has fewer below-target surfaces, it combines a higher mean coverage of 94.76% with substantially wider bands. Score B has a mean coverage of 91.34%, lower variation across surfaces, and considerably narrower bands, providing a closer balance between coverage and sharpness.

5.4.4 Summary of Score Comparison

Across projection behaviour, band width, and empirical coverage, Score B provides the more balanced result. Score A retains residual calendar and butterfly violations under the grid-based routine and produces substantially wider bands. Score B reaches zero residual violations under the same routine, produces narrower bands, and achieves mean coverage closer to the nominal target with lower variation across surfaces. The two outcomes reflect their different constructions: Score B forms bands directly from Vega-weighted IV residuals, whereas Score A calibrates spread-normalised price residuals and then maps the resulting price bands into IV space. Score B is therefore the recommended construction for this framework, and all detailed results reported in Sections 5.5 and 5.6 use Score B exclusively.

5.5 Coverage Results

This section examines the empirical marginal coverage of the Score B framework in detail across the 41 held-out evaluation surfaces. Coverage is the fraction of market implied volatility quotes that fall within the constructed band at each evaluation surface, as defined in Section 3.6. Two variants are reported at the same post-projection stage: the global construction uses a single scalar quantile from the rolling window, whereas the Mondrian construction uses per-bin quantiles from the same window. Both use the same $K = 5$ rolling window and the same projection routine; the difference is the quantile assignment.

5.5.1 Per-Surface Coverage

Table 5.5 reports the matched post-projection global and Mondrian coverage for Score B on each evaluation surface.

Idx	Date	Global	Mondrian	Idx	Date	Global	Mondrian
15	2026-01-29	90.57%	91.44%	36	2026-06-23	89.01%	92.62%
16	2026-01-30	90.84%	92.28%	37	2026-06-24	88.27%	90.68%
17	2026-02-11	91.17%	91.91%	38	2026-06-26	91.97%	94.76%
18	2026-02-14	91.17%	91.96%	39	2026-06-27	88.31%	89.60%
19	2026-02-17	88.94%	91.52%	40	2026-07-02	91.63%	91.80%
20	2026-03-03	90.74%	92.15%	41	2026-07-03	88.93%	87.57%
21	2026-04-10	90.17%	92.03%	42	2026-07-07	89.07%	90.77%
22	2026-04-16	92.47%	93.40%	43	2026-07-08	87.48%	90.64%
23	2026-04-17	90.93%	92.23%	44	2026-07-15	91.59%	94.45%
24	2026-05-08	94.53%	95.60%	45	2026-07-17	90.20%	89.13%
25	2026-05-13	89.81%	90.98%	46	2026-07-18	89.95%	90.92%
26	2026-05-14	90.36%	91.72%	47	2026-07-30	89.53%	89.87%
27	2026-05-15	93.51%	94.88%	48	2026-07-31	90.12%	89.72%
28	2026-05-18	90.71%	90.75%	49	2026-08-04	88.61%	90.41%
29	2026-05-19	89.96%	91.66%	50	2026-08-05	90.82%	91.19%
30	2026-05-20	91.18%	92.72%	51	2026-08-06	89.52%	88.66%
31	2026-05-21	89.54%	91.33%	52	2026-08-07	91.24%	91.51%
32	2026-05-22	92.22%	93.89%	53	2026-08-08	91.31%	92.37%
33	2026-06-04	87.91%	89.92%	54	2026-08-11	86.12%	87.56%
34	2026-06-05	89.04%	90.78%	55	2026-08-15	88.77%	89.07%
35	2026-06-16	91.38%	88.35%				
Mean		90.23%	91.34%				
Std		1.62%	1.86%				

Table 5.5: Matched post-projection Score B coverage on all 41 evaluation surfaces ($1 - \alpha = 90\%$).

Over the full 41-surface evaluation period, the mean Mondrian coverage is 91.34%, above the 90% nominal target. Its standard deviation is 1.86 percentage points. The minimum Mondrian coverage is 87.56% (surface 54, 11 August

2026), and the maximum is 95.60% (surface 24, 8 May 2026). Mondrian coverage falls below the 90% target on 10 of the 41 evaluation surfaces; these cases are considered in Section 5.5.2.

5.5.2 Contribution of Mondrian Spatial Binning

The global coverage column of Table 5.5 provides the comparison without spatial binning while retaining the same rolling window and projection stage. Its mean coverage is 90.23%, compared with 91.34% for Mondrian coverage.

Mondrian spatial binning raises mean post-projection coverage by 1.10 percentage points. The improvement is visible on several individual surfaces. For example, coverage increases from 89.01% to 92.62% on surface 36 and from 87.48% to 90.64% on surface 43. Across the full evaluation period, 11 surfaces have global coverage below 90% but Mondrian coverage at or above 90%. These results show the benefit of assigning calibration quantiles by surface region rather than applying one value across the full domain.

The ten surfaces where Mondrian coverage falls below 90% are surfaces 33, 35, 39, 41, 45, 47, 48, 51, 54, and 55. Global coverage reaches at least 90% on three of these surfaces (35, 45, and 48), showing that Mondrian binning improves average coverage but does not increase coverage on every individual surface. At each walk-forward step, coverage is evaluated using the preceding rolling window; the current surface is added only after its evaluation and can then contribute to the next window. The window therefore contains the five most recently processed surfaces without using future information. The remaining temporal variation and its relation to the conformal assumptions are discussed further in Chapter 6.

5.6 Sharpness

Coverage is a necessary condition for a useful prediction band but not a sufficient one: a very wide band can achieve high coverage while providing little information about the likely range of the surface. Sharpness measures the width of the prediction band. When coverage is comparable, narrower bands are generally more informative. This section reports the IV band widths produced by the Score B framework across all 41 evaluation surfaces.

5.6.1 Overall Band Width

Averaging the per-surface mean IV band widths across the 41 evaluation surfaces gives 0.0273 IV units. The corresponding average of the per-surface median widths is 0.0262 IV units. These figures reflect the full pipeline, comprising Mondrian quantile assignment followed by Vega-IV band construction and no-arbitrage projection, and are measured on the scattered evaluation quotes after the projected grid values have been mapped back to the original (ρ, z) coordinates.

The GNO point-prediction MAE is 0.006789 IV units. The mean half-width of 0.0137 IV units is therefore approximately twice the average prediction error. Together with the mean Mondrian coverage of 91.34%, this provides a direct summary of the observed relationship between coverage and band width.

5.6.2 Spatial Variation of Band Width

The Mondrian construction permits spatially varying band widths by assigning calibration quantiles according to the (ρ, z) bin containing each point. At the final calibration surface, the bin quantiles range from 0.011950 to 0.040613 (Table 5.1), a factor of approximately 3.4 between the lowest and highest calibration thresholds. This ratio describes variation in the thresholds rather than a direct factor-of-3.4 range in the final widths, which also depend on the local Vega weight and the subsequent projection. The width statistics reported above are averages of per-surface summaries computed over the evaluation quotes. As shown in Section 5.5.2, this spatial assignment raises mean post-projection coverage relative to the global construction, although it does not increase coverage on every individual surface.

5.6.3 Sharpness Relative to Score A

As established in Section 5.4.2, Score A produces bands with a mean width of 0.1491 IV units, 5.5 times wider than Score B's 0.0273. Score A achieves mean coverage of 94.76%, whereas Score B achieves 91.34%, closer to the 90% target. The comparison therefore shows a clear coverage–sharpness trade-off: Score A obtains higher coverage with substantially broader bands, while Score B provides considerably narrower bands with mean coverage closer to the nominal level. This difference reflects their distinct constructions: Score A calibrates spread-normalised price residuals and maps the resulting price bands into IV space, whereas Score B constructs the bands from Vega-weighted IV residuals.

5.7 Summary

This chapter has reported the empirical evaluation of the conformal uncertainty quantification framework on 56 successfully collected S&P 500 options surface snapshots with collection dates spanning December 2025 to August 2026, with 15 surfaces used for initial calibration and 41 evaluated sequentially in walk-forward order.

The calibration results in Section 5.2 showed that the Score B global quantile remained within a relatively narrow range across the calibration period. It moved from 0.01623 after the first surface to 0.01769 after the final calibration surface, with intermediate values ranging from 0.01568 to 0.01781 as the rolling window advanced. By the end of calibration, the Mondrian bin quantiles spanned a factor-of-3.4 range, showing that the Score B error distribution varies across regions of the implied volatility surface and that the 4×4 binning represents this spatial variation.

The no-arbitrage projection results in Section 5.3 showed stronger performance for Score B under the implemented grid-based conditions. For Score B, the combined lower-and-upper calendar and butterfly count decreased from 59,414 to zero across the 41 evaluation surfaces. For Score A, it decreased from 82,190 to 43,471, a reduction of 47.1%, while lower–upper crossings remained zero for both scores. Together with the substantially greater Score A band width, these results support the use of Score B as the primary construction and the retention of Score A as a comparative baseline.

The Score A versus Score B comparison in Section 5.4 reinforced this conclusion across two further dimensions. Score A's mean Mondrian coverage of 94.76% over 41 evaluation surfaces comes with a standard deviation of 3.31 percentage points and substantially wider bands. Score B's mean coverage of 91.34%, which is closer to the 90% target, has a standard deviation of 1.86 percentage points and considerably narrower bands.

The detailed Score B coverage results in Section 5.5 showed that the Mondrian construction achieves a mean coverage of 91.34% across the 41 evaluation surfaces. On 11 surfaces where global coverage is below the 90% target, Mondrian binning raises coverage to at least 90%. It does not improve every individual surface, and ten surfaces from June to August 2026 remain below 90% after Mondrian assignment. This remaining temporal variation is considered further in Chapter 6.

The sharpness results in Section 5.6 showed that Score B produces bands with a mean IV width of 0.0273 units and a corresponding mean half-width of approximately 0.0137 units. The mean half-width is approximately twice the GNO's mean absolute error of 0.006789 IV units. Score B bands are 5.5 times narrower than Score A bands in mean IV width while maintaining mean empirical coverage above the nominal target.

Taken together, the empirical results show that the proposed framework, comprising Score B nonconformity scores, $K = 5$ rolling window calibration, 4×4 Mondrian spatial binning with $\lambda = 0.3$ shrinkage, Vega-IV band construction, and no-arbitrage projection, produces spatially adaptive bands with mean coverage above the nominal target and substantially narrower widths than Score A. For Score B, the projection reduces the residual counts for the implemented calendar and butterfly conditions to zero on all evaluation surfaces. Chapter 6 interprets these findings in the context of the exchangeability assumption, their practical implications, and the limitations of the current framework.

Chapter 6

Discussion

This chapter interprets the empirical findings of Chapter 5 in broader context. Section 6.1 discusses the exchangeability assumption underlying conformal coverage and its relevance to the temporally ordered SPX options data. Section 6.2 considers the practical interpretation of the resulting band surfaces. Section 6.3 discusses the limitations of the current study, and Section 6.4 outlines focused directions for future research. Section 6.5 summarises the main points.

6.1 The Exchangeability Assumption in Practice

The standard finite-sample coverage guarantee of conformal prediction relies on exchangeability of the calibration and test nonconformity scores, meaning that their joint distribution is invariant to permutations of their indices [19, 2]. Independent and identically distributed observations provide a common sufficient condition, but exchangeability is the more general requirement. In the present application, option surfaces are ordered in time and the quotes within a surface are related through the same market state. Exact exchangeability is therefore not assumed. The rolling procedure is used as a practical adaptation mechanism, and its coverage is evaluated empirically in Chapter 5.

6.1.1 Serial Dependence and the Rolling Window

The rolling window restricts calibration to the $K = 5$ most recently processed surfaces. This keeps the calibration information temporally close to the surface being evaluated and prevents distant observations from remaining in the window indefinitely. The window counts available surfaces rather than elapsed calendar days, which is appropriate for the irregular sequence of successfully obtained snapshots used in this thesis. However, restricting the window does not by itself establish exchangeability or restore the formal finite-sample guarantee under temporal dependence. It instead provides a simple way to make the conformal quantile responsive to recently observed errors, consistent with the motivation for giving recent observations greater relevance under distribution drift [3].

The Score B global quantile remains within a relatively narrow range during the initial calibration period. It moves from 0.01623 after the first surface to 0.01769 after the final calibration surface, with intermediate values between 0.01568 and 0.01781. This shows that the rolling estimate changes as surfaces enter and leave the window without being dominated by a single calibration date. The calibration behaviour alone cannot verify exchangeability, but it shows that the selected window produces a usable and gradually adapting threshold for the subsequent walk-forward evaluation.

6.1.2 Temporal Variation in Coverage

Across the 41 evaluation surfaces, post-projection global coverage is below the 90% target on 18 surfaces, while Mondrian coverage is below 90% on 10 surfaces. The shortfalls are not confined to one part of the evaluation period.

For example, surface 19 has global coverage of 88.94%, which increases to 91.52% with Mondrian assignment, whereas surface 54 remains below target at 86.12% globally and 87.56% with Mondrian assignment. These results show that the relevance of the preceding rolling window varies over time.

At each walk-forward step, the current surface is evaluated before its scores are added to the window. The procedure can therefore adapt after a new error pattern has been observed, but it cannot anticipate a change that is not represented in the preceding surfaces. The available data do not include a separate analysis of market regimes or other covariates that would identify the cause of each coverage shortfall. The results are therefore interpreted as empirical temporal variation rather than attributed to a specific market event. Barber et al. [3] quantify how departures from exchangeability can reduce coverage through bounds based on differences between original and index-swapped data or residual distributions. This supports the cautious interpretation adopted here: the rolling method is useful empirically, but the standard exchangeable-data guarantee is not claimed for the time-dependent evaluation.

6.1.3 Contribution of Mondrian Binning

Mondrian binning raises mean post-projection coverage from 90.23% for the global construction to 91.34%. On 11 surfaces where global coverage is below 90%, the Mondrian construction raises coverage to at least the target. It does not improve every surface, and ten surfaces remain below 90%. The effect is therefore beneficial on average rather than uniform across time.

The calibration results help explain this improvement. At the final calibration surface, the bin-specific Score B quantiles range from 0.011950 to 0.040613, showing substantial spatial variation around the global quantile of 0.017689. Assigning these thresholds by (ρ, z) region allows the band width to respond to the observed spatial distribution of errors. Mondrian conformal prediction is motivated by calibration within predefined categories under exchangeability [20, 19]. In this thesis, the rolling window and shrinkage toward the global quantile make the construction an empirical spatial adaptation rather than a claim of exact conditional coverage.

6.2 Practical Implications

The framework produces complete lower and upper surfaces around the smooth GNO estimate over the modelled domain. The conformal calibration is performed using errors at observed quote locations, after which interpolation and projection provide the surface representation. Empirical coverage is evaluated at the observed quotes, where market implied volatility is available for comparison. This combines a surface-level output with a directly measurable quote-level evaluation.

The width of the band provides an interpretable indication of uncertainty around the GNO surface. For Score B, the mean IV width is 0.0273 and the corresponding mean half-width is approximately 0.0137, compared with a GNO mean absolute error of 0.006789. The bands may therefore support model monitoring by showing the range around the point surface that is consistent with recent prediction errors. Wider regions indicate where the calibrated uncertainty is larger, while narrower regions indicate greater agreement with the recent calibration window. The bands should be interpreted as model uncertainty estimates rather than as direct trading or hedging recommendations.

The comparison of the two scores supports the use of Score B for the present application. Score A achieves higher mean coverage of 94.76%, but its mean IV width of 0.1491 is approximately 5.5 times the Score B width and its projected surfaces retain residual calendar and butterfly violations. Score B achieves mean Mondrian coverage of 91.34%, reaches zero residual counts for the implemented calendar and butterfly conditions on all evaluation surfaces, and has zero lower–upper crossings. It therefore provides the more balanced combination of coverage, band width, and financial shape adjustment in this dataset.

The projection operates on the complete lower and upper surfaces rather than adjusting isolated quote intervals. For Score B, the resulting surfaces satisfy the two implemented discrete conditions on the evaluation grids while preserving their lower–upper ordering at the observed quotes. This makes the projected bands easier to use as coherent surface objects in subsequent analysis.

6.3 Limitations

The empirical study uses one underlying asset, one pre-trained GNO checkpoint, and 56 successfully collected SPX option surface snapshots. The 41-surface evaluation covers collection dates from 29 January to 15 August 2026, but the sequence is irregular because provider-side API errors and timeouts prevented successful retrieval on some scheduled dates. The retrieval script records the collection date in the saved schema, rather than a separate exchange-provided quote timestamp. The dates therefore provide a chronological index for the successfully collected chains and should not be interpreted as a complete sample of trading days. The results provide evidence for this dataset and period and should not be assumed to transfer unchanged to other underlyings, models, or market periods.

The rolling window size $K = 5$, the 4×4 Mondrian partition, and shrinkage parameter $\lambda = 0.3$ are held fixed throughout the reported experiments. Alternative configurations were considered during preliminary trials, and the selected values provided the best overall balance of coverage, band width, and stability for the available data. The thesis therefore uses this configuration for the comparison of global and Mondrian calibration and of Score A and Score B. A systematic sensitivity analysis over all possible window lengths, bin counts, and shrinkage values remains outside the scope of the present study.

Coverage is measured over the observed quotes on each evaluation surface and then summarised across surfaces. Quotes within the same surface are cross-sectionally related, and regions containing more observed quotes contribute more observations to that surface's coverage value. Together with temporal dependence between surfaces, this reinforces the use of empirical coverage as the main assessment rather than treating the standard exchangeability result as exact for this dataset.

The calendar and butterfly adjustments are implemented as discrete conditions on a finite (ρ, z) grid. The calendar step operates across maturities at fixed z , matching the transformed coordinate representation used by the implementation. A fixed- k calendar calculation is retained as a supplementary diagnostic rather than an enforced condition. Similarly, the butterfly step applies the implemented chord condition across z at each maturity. These choices provide a stable and transparent surface adjustment, while alternative grid formulations could be considered in a broader implementation.

The lower and upper surfaces are projected independently. No lower–upper crossings are observed in the reported experiments, but the independent procedure does not impose their joint ordering as part of a single optimisation problem. The zero crossing count is therefore an empirical result of the current evaluation.

6.4 Future Research

A first extension would evaluate the framework on a longer sequence of SPX surfaces and on additional index or single-name option datasets. This would show whether the relative performance of Score A and Score B, the selected band widths, and the behaviour of the rolling calibration remain similar across other assets and periods.

A second extension would examine sensitivity to the main calibration choices, including the rolling window length, the number of Mondrian bins, and the shrinkage parameter. Weighted or adaptive conformal methods could also be considered if stronger adaptation to temporal distribution drift is required [3]. These extensions would complement the current fixed configuration rather than alter the conclusions obtained from the reported experiment.

A further implementation extension could formulate the calendar adjustment directly on a fixed- k grid and compare it with the current fixed- z construction. The lower and upper surfaces could also be projected jointly so that their ordering is imposed directly. Finally, comparisons with additional volatility-surface or uncertainty baselines would provide a broader assessment of computational and statistical trade-offs.

6.5 Summary

The empirical results show that rolling calibration and Mondrian spatial binning provide a practical uncertainty layer around the pre-trained GNO, although the temporal structure of the data means that the standard exchangeability guarantee is not treated as exact. Score B provides the preferred balance in the reported experiment: its mean coverage is above the nominal target, its bands are substantially narrower than those of Score A, and the implemented calendar

and butterfly conditions have zero residual counts after projection on all evaluation surfaces. The limitations of the data scope, fixed calibration choices, and discrete projection define clear extensions without changing the main conclusion of the thesis.

Chapter 7

Conclusion

Bibliography

- [1] Damien Ackerer, Natasa Tagasovska, and Thibault Vatter. “Deep smoothing of the implied volatility surface.” In: *Advances in Neural Information Processing Systems* 33 (2020), pp. 11552–11563. URL: <https://arxiv.org/abs/1906.05065>.
- [2] Anastasios N. Angelopoulos and Stephen Bates. “Conformal Prediction: A Gentle Introduction.” In: *Foundations and Trends in Machine Learning* 16.4 (2023), pp. 494–591. DOI: 10.1561/2200000101. URL: <https://doi.org/10.1561/2200000101>.
- [3] Rina Foygel Barber et al. “Conformal prediction beyond exchangeability.” In: *The Annals of Statistics* 51.2 (2023), pp. 816–845. DOI: 10.1214/23-AOS2276. URL: <https://arxiv.org/abs/2202.13415>.
- [4] Maxime Bergeron et al. “Variational autoencoders: A hands-off approach to volatility.” In: *Journal of Financial Data Science* 4.2 (2022), pp. 125–138. DOI: 10.3905/jfds.2022.1.093. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3827447.
- [5] Fischer Black and Myron Scholes. “The pricing of options and corporate liabilities.” In: *Journal of Political Economy* 81.3 (1973), pp. 637–654. DOI: 10.1086/260062. URL: https://www.cs.princeton.edu/courses/archive/fall09/cos323/papers/black_scholes73.pdf.
- [6] Alejandro Canete. “Market Implied Conformal Volatility Intervals.” In: *Proceedings of the Twelfth Symposium on Conformal and Probabilistic Prediction with Applications*. Vol. 204. Proceedings of Machine Learning Research. 2023, pp. 89–99. URL: <https://proceedings.mlr.press/v204/canete23a.html>.
- [7] Cboe Global Markets, Inc. *Cboe Global Markets Reports Trading Volume for December and Full Year 2025*. Press Release. Accessed: May 19, 2026. Jan. 2026. URL: <https://ir.cboe.com/news/news-details/2026/Cboe-Global-Markets-Reports-Trading-Volume-for-December-and-Full-Year-2025/default.aspx>.
- [8] Yarin Gal and Zoubin Ghahramani. “Dropout as a Bayesian approximation: Representing model uncertainty in deep learning.” In: *Proceedings of the 33rd International Conference on Machine Learning*. Vol. 48. 2016, pp. 1050–1059. URL: <https://arxiv.org/abs/1506.02142>.
- [9] Jim Gatheral and Antoine Jacquier. “Arbitrage-free SVI volatility surfaces.” In: *Quantitative Finance* 14.1 (2014), pp. 59–71. DOI: 10.1080/14697688.2013.819986. URL: <https://arxiv.org/abs/1204.0646>.
- [10] Lukas Gonon, Antoine Jacquier, and Ruben Wiedemann. “Operator Deep Smoothing for Implied Volatility.” In: *arXiv preprint arXiv:2406.11520v1* (2024). URL: <https://arxiv.org/abs/2406.11520v1>.
- [11] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. “Simple and scalable predictive uncertainty estimation using deep ensembles.” In: *Advances in Neural Information Processing Systems* 30 (2017). URL: <https://arxiv.org/abs/1612.01474>.
- [12] Jing Lei et al. “Distribution-free predictive inference for regression.” In: *Journal of the American Statistical Association* 113.523 (2018), pp. 1094–1111. DOI: 10.1080/01621459.2017.1307116. URL: <https://arxiv.org/abs/1604.04173>.
- [13] Zongyi Li et al. “Neural operator: Graph kernel network for partial differential equations.” In: *arXiv preprint arXiv:2003.03485* (2020). URL: <https://arxiv.org/abs/2003.03485>.

- [14] Ziqi Ma et al. "Calibrated Uncertainty Quantification for Operator Learning via Conformal Prediction." In: *Transactions on Machine Learning Research* (2024). URL: <https://openreview.net/forum?id=cGpegxy12T>.
- [15] OpenBB. *Options Chains: Python Reference*. OpenBB Documentation. Accessed: August 30, 2026. 2026. URL: <https://docs.openbb.co/odp/python/reference/derivatives/options/chains>.
- [16] Harris Papadopoulos et al. "Inductive confidence machines for regression." In: *Machine Learning: ECML 2002*. Springer, 2002, pp. 345–356. DOI: 10.1007/3-540-36755-1_29. URL: https://link.springer.com/content/pdf/10.1007/3-540-36755-1_29.pdf.
- [17] Yaniv Romano, Evan Patterson, and Emmanuel Candès. "Conformalized quantile regression." In: *Advances in Neural Information Processing Systems* 32 (2019). URL: <https://arxiv.org/abs/1905.03222>.
- [18] Johannes Ruf and Weiguan Wang. "Hedging with Linear Regressions and Neural Networks." In: *Journal of Business & Economic Statistics* 40.4 (2022), pp. 1442–1454. DOI: 10.1080/07350015.2021.1931241. URL: <https://doi.org/10.1080/07350015.2021.1931241>.
- [19] Glenn Shafer and Vladimir Vovk. "A tutorial on conformal prediction." In: *Journal of Machine Learning Research* 9 (2008), pp. 371–421. URL: <https://jmlr.org/papers/v9/shafer08a.html>.
- [20] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. New York: Springer, 2005. DOI: 10.1007/b106715. URL: <https://link.springer.com/book/10.1007/b106715>.
- [21] Magnus Wiese et al. "Deep Hedging: Learning to Simulate Equity Option Markets." In: *arXiv preprint arXiv:1911.01700* (2019). URL: <https://arxiv.org/abs/1911.01700>.